Physics-Aware Multi-Objective Scheduling for an Agile SAR Satellite: Balancing Profit, NESZ-Based Radiometric Ordering, and Geometric Resolution

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Abstract

In agile SAR scheduling the choice of observation time inside a visibility window sets the image quality, so scheduling and imaging physics act as one decision. We formulate single-satellite Stripmap scheduling as a three-objective problem that optimises coverage profit, geometric resolution, and noise-equivalent sigma-zero (NESZ) radiometric quality jointly with a non-dominated sorting genetic algorithm III (NSGA-III), and we measure where the formulation pays off. Across 200 simulated scenarios spanning 20 to 500 targets in a two-orbit horizon, sparse request sets reward re-optimisation over the greedy coverage baseline: +33.7% geometric-index quality at a 14% coverage cost and +40% radiometric-index quality at a 15% cost (the squint-minimising greedy heuristic reaches +45% at a 49% cost), with a controlled +4.9% third-objective margin under a matched radiometric-aware selection rule (+0.05%, $p = 0.86$, radiometric-blind).

Two measured boundaries delimit that gain. The advantage contracts as the squint envelope narrows, from +28% to +2.3% in geometric quality when the cap tightens from 45° to 15°, and it vanishes beyond $N \approx 150$ (logistic transition midpoint $N_{50} \approx 51$, CI $[37, 64]$), where the multi-objective fronts converge to the greedy coverage anchor with quality margins below 2% under greedy hot-start initialization. The geometric and radiometric objectives conflict most inside sparse schedules ($r \approx -0.62$, close to the -0.60 correlation two independent angles would produce), and coverage pressure weakens that

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conflict to between -0.13 and -0.23 at high density, which locates the loss of value in the schedule geometry and not in a redundancy between objectives. A four-variant objective ablation over 1,200 further runs traces these gains to the squint channel rather than to objective redundancy. Together the boundaries give planners two quantified operating limits: squint-envelope contraction and a density threshold, plus an optional operating point, the knee point of that trade-off.

Keywords: agile SAR, satellite scheduling, multi-objective optimization, NSGA-III, noise-equivalent sigma-zero (NESZ, relative ordering index), geometric resolution

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1. Introduction

Agile Earth-observation satellites equipped with Synthetic Aperture Radar (SAR) are increasingly central to time-critical monitoring missions, from disaster response to maritime surveillance. Their agility (the ability to slew the antenna beam away from the nadir track) enables a single satellite to image far more targets per orbit than a fixed-pointing platform (Curlander and McDonough, 1991). This flexibility, however, transforms scheduling from a simple feasibility check into a combinatorial optimization problem: for N targets each with a finite visibility window, the scheduler must select a subset and assign observation times that satisfy attitude transition constraints while maximizing some measure of mission value.

Throughout this paper, the setting is a *single-satellite, two-orbit-horizon* problem: one satellite plans observations over a two-orbit scheduling horizon (~ 200 min), with at most one observation per target.

In conventional satellite scheduling, observation quality is commonly treated as a fixed attribute of each pre-computed opportunity. Under that treatment the scheduling problem reduces to maximizing the number or priority-weighted sum of observed targets subject to temporal feasibility, an assumption that underpins most existing satellite scheduling work, from mixed-integer programming formulations to deep reinforcement learning approaches (Chen et al., 2025).

SAR imaging departs fundamentally from this assumption when the scheduler can choose the acquisition time, because image quality depends on the radar geometry at the moment of acquisition: the ground-range resolution degrades with incidence angle ($\rho_{\mathrm{rg}} \propto 1/\sin\theta$), the azimuth resolution

broadens with squint angle ($\rho_{\text{az}} \propto 1/\cos\psi_{\text{sq}}$), and the radiometric sensitivity (Noise-Equivalent Sigma Zero, NESZ) deteriorates with slant range as R^3 (Cumming and Wong, 2005). When the scheduler can choose *when* within a visibility window to observe each target (the defining feature of agile satellites), it simultaneously chooses the imaging quality at which each target is acquired. Despite this tight coupling between scheduling and imaging physics, with few exceptions (Kramer and Miller, 2025), existing agile SAR scheduling methods either reduce SAR to a coverage-only objective or apply post hoc quality checks on schedules produced by coverage-maximizing solvers.

The problem is NP-hard by reduction from the Orienteering Problem with Time-Dependent Profits (Peng et al., 2019), precluding polynomial-time exact algorithms for all but the smallest instances. The SAR literature has well-established models for NESZ and geometric resolution as functions of observation geometry (Curlander and McDonough, 1991; Cumming and Wong, 2005). We therefore formulate SAR scheduling with separate, physics-derived geometric and radiometric quality objectives within a multi-objective evolutionary algorithm (MOEA) framework.

We formulate a three-objective agile SAR scheduling problem in which the scheduler optimizes (1) total coverage profit, (2) mean geometric image quality (ground-range and azimuth resolution), and (3) mean NESZ radiometric quality simultaneously (denoted f_1^*, f_2, and f_3 below, § 3.2). We compare five solver configurations spanning the methodological spectrum: two greedy heuristics (the profit-only greedy baseline G-BL and one squint-minimized variant G-SM), a single-objective genetic algorithm with greedy hot-start (GA-P-BL, a control that returns the greedy schedule whenever it fails to improve profit, so it is never worse than G-BL), and two NSGA-III variants (MOEA-2 and MOEA-3) with two and three objectives.

The specific contributions of the paper are:

1. A three-objective formulation of single-satellite agile SAR scheduling that decomposes image quality into closed-form NESZ ($\propto R^3$) and geometric quality ($\sin\theta\cos\psi_{\text{sq}}$, the dimensionless inverse of the ground pixel area since $\rho_{\text{rg}} \propto 1/\sin\theta$ and $\rho_{\text{az}} \propto 1/\cos\psi_{\text{sq}}$) objectives inside NSGA-III, using mean per-task quality metrics that decouple quality from the number of scheduled tasks. This separates the physics from the coarse visibility composite of Kramer and Miller (2025) and the generic image-quality composite of Wei et al. (2025), and enables a systematic quality–coverage comparison across two greedy heuristics,

a single-objective GA control, and two NSGA-III variants on 200 scenarios in four scale classes;
2. A measured operating envelope for that formulation: the size of the sparse-regime gain (+33.7% in f_2 at $N = 20$), its contraction under a tightening squint cap (from +28% to +2.3% as the cap narrows from 45° to 15°), and the density threshold beyond which the multi-objective variants converge to the greedy coverage anchor ($N_{50} \approx 51$; negligible advantage beyond $N = 150$) (§6.2, §6.7);
3. An anchor-independent decomposition of the f_2–f_3 coupling that separates the trade-off carried by raw visibility geometry from the within-schedule residual (§6.4): measured across the density classes, the trade-off is mild over raw visibility geometry ($r \approx -0.36$ against an independent-angle null of $r \approx -0.60$), is restored to null strength inside sparse schedules (-0.62) by squint-aware steering, and dilutes to -0.13 to -0.23 under coverage pressure at high density;
4. A controlled ablation (four objective-function variants over six scenario classes, 300 scenarios per variant) and initialization controls showing that the third objective's measurable value is concentrated in sparse schedules and flows through the squint channel, that solver outcomes become invariant to the choice of physical objectives at high target density (where the objective is retained for quality-assessment completeness rather than incremental gain), and that coverage convergence is conditional on greedy-baseline (G-BL) hot-start initialization (§6.5, §6.6).

2. Related Work

We examine three lines of research that our study connects: satellite scheduling (2.1), SAR physical quality modeling (2.2), and multi-objective evolutionary optimization (2.3).

2.1. Satellite Scheduling

Satellite scheduling is surveyed in depth by Ferrari et al. (2025) (over 200 publications across exact, heuristic, metaheuristic, and learning methods). Mixed-integer linear programming is exact on small instances but does not scale, motivating genetic algorithms, simulated annealing, tabu search (Wang et al., 2019; Chen et al., 2019), a Lagrangian-relaxation heuristic for multi-satellite planning (Wang et al., 2026), and constraint program-

ming for the super-agile problem (Caleiras et al., 2026); Lemaître et al. (2002) and Bianchessi et al. (2007) established the agile problem's time-dependent transition structure, and recent work packs more observations per orbit via free start times within visibility windows (Peng et al., 2019; Chen et al., 2024). A fast-growing parallel line is learning-based: Liu et al. (2024) and Zhou et al. (2025) apply deep reinforcement learning to agile optical satellite scheduling, Jacquet et al. (2024) combine graph networks with Monte Carlo tree search, and Herrmann and Schaub (2023) apply RL on-board without imaging quality; for high-density AEOS, Wu et al. (2026) report convergence challenges echoing the density-driven tightening we analyze in § 6.2. Most prior work treats imaging quality as fixed at window generation, whereas SAR quality varies continuously with acquisition geometry and is directly optimizable inside the scheduling window. The SAR-specific branch applies multi-objective evolutionary algorithms (MOEAs) to this gap, but with quality as a scalar or proxy: Chang et al. (2022) pair large-neighborhood search with NSGA-II/III over a three-objective operational model (ground-target loss, invalid observations, sensor activations; the closest prior work to ours, but without imaging quality); Wei et al. (2025) use a neural-policy MOEA over a composite score; Kramer and Miller (2025) embed a quality proxy for constellations; Yao et al. (2024) separate allocation from detailed planning; Chang and Zhou (2025) attach an attitude-angle proxy; Wu et al. (2019) use NSGA-III with coverage and timeliness objectives. The common limitation of this prior work is that SAR physical quality metrics such as noise-equivalent sigma zero (NESZ), azimuth resolution, and ground-range resolution are either absent or approximated by coarse proxies (e.g., penalizing extreme off-nadir angles). To our knowledge, no existing SAR scheduling method treats continuous, physics-derived quality functions as decomposed optimization objectives in their own right: the closest approaches embed aggregated quality proxies rather than separate geometric and radiometric objectives.

2.2. SAR Physical Quality Modeling

The SAR remote sensing literature provides well-established analytical models linking observation geometry to image quality. Curlander and McDonough (1991), Cumming and Wong (2005), and Moreira et al. (2026) give detailed treatments of SAR physics, including the two quality dimensions relevant to scheduling: geometric resolution and radiometric sensitivity. Geometric resolution decomposes into ground-range resolution $\rho_{\mathrm{rg}} = c/(2B\sin\theta)$,

where c is the speed of light, B the radar bandwidth, and θ the incidence angle, and azimuth resolution $\rho_{\text{az}} = D/(2\cos\psi_{\text{sq}})$, where D is the antenna length and ψ_{sq} the squint angle. In operational SAR, amplitude weighting widens ρ_{az} by 10–20% (Curlander and McDonough, 1991), but since this is platform-constant, the unweighted form preserves the relative ordering for optimization. The two resolution components respond oppositely to departure from the zero-Doppler geometry ($\psi_{\text{sq}} = 0$, θ minimal): ρ_{az} worsens as $|\psi_{\text{sq}}|$ grows, whereas ρ_{rg} is coarsest at the minimal incidence of closest approach and improves as θ rises away from it, the origin of the θ-axis trade-off formalized in § 3.2. Radiometric sensitivity, measured by the Noise-Equivalent Sigma Zero (NESZ), deteriorates with the cube of slant range: NESZ $\propto R^3$ for Stripmap SAR after azimuth compression gain (Curlander and McDonough, 1991; Moreira et al., 2026). The specific range decomposition and ordering-index approximation used in this work are derived in § 3.2. Our study connects these two domains by embedding continuous quality functions directly into the optimization objective space.

2.3. Multi-Objective Evolutionary Optimization

Multi-objective evolutionary algorithms are a standard toolkit for problems with competing objectives. The NSGA-III algorithm (Deb and Jain, 2014) extends the NSGA-II framework (Deb et al., 2002) to many-objective problems through reference-point-based diversity preservation. The hypervolume (HV) indicator (Zitzler and Thiele, 1999) is a standard Pareto-front comparison tool; its interpretational caveats for mixing single- and multi-objective solvers are handled in § 5.3 and § 6.4. For this study's use, three caveats matter. First, with origin-anchored reference boxes HV is insensitive to cardinality—dominated points add no volume, so one extreme point can out-volume an entire front. Second, it is blind to decision-space diversity, making front-ranking claims fragile when single-point greedy solvers and multi-point NSGA-III fronts are mixed. Third, these properties rule out HV as a standalone quality judgment, so we pair it with per-objective effect sizes and report it as a secondary diagnostic with explicit point-count controls. Our NSGA-III formulation makes these trade-offs explicit by placing coverage profit, geometric quality, and radiometric quality in a unified multi-objective framework.

3. Problem Formulation

3.1. System Model

We consider a single agile SAR satellite in a circular orbit at altitude H, operating in Stripmap mode, which provides the widest continuous swath with uniform quality and analytically tractable NESZ and resolution expressions (Curlander and McDonough, 1991). A problem instance $\mathcal{I} = (\mathcal{T}, \mathcal{P}, \mathcal{O})$ consists of a set of N ground targets $\mathcal{T}$, a satellite platform $\mathcal{P}$, and a SAR imaging model $\mathcal{O}$.

Task Set.. Each task $i \in \{1, \ldots, N\}$ is specified by its geographic coordinates $\mathbf{r}_i = (\phi_i, \lambda_i)$, a priority weight $p_i \in \mathbb{R}^+$, a visibility window $\mathcal{W}_i = [a_i, b_i]$ during which the satellite can observe the target, and a fixed observation duration d_i.

Satellite Platform.. The satellite is characterized by its maximum slew rate $\omega_{\max}$ (a conservative single-axis bound on three-axis attitude maneuvers), settling time τ_{settle} after each maneuver, per-orbit energy budget $E_{\max}$ and memory budget $M_{\max}$, and per-observation energy e_i and memory m_i consumption.

SAR Observables.. Given an observation time t_i, the satellite-to-target line-of-sight (LOS) vector $\mathbf{l}(t_i)$ in the Local-Vertical Local-Horizontal (LVLH) frame is fully determined by the satellite's orbital state. From $\mathbf{l}(t_i)$ we derive four geometric quantities deterministically: slant range $R = |\mathbf{l}|$, off-nadir angle $\xi = \arctan 2(\sqrt{l_x^2 + l_y^2},\, l_z)$, incidence angle θ (via the Earth-curvature-corrected mapping $\theta = \arcsin\left(\frac{R_e+H}{R_e} \sin \xi\right)$ where R_e is Earth's mean radius), and squint angle $\psi_{\text{sq}} = \arcsin(|l_x|/R)$ (LVLH x-axis along-track, z-axis nadir-pointing), defined as the angular deviation of the LOS vector from the zero-Doppler plane (Curlander and McDonough, 1991). For the objectives we also use the elevation-plane incidence angle θ_{elev}, defined through the identity $\cos\theta_{\text{elev}} = \cos\xi / \cos\psi_{\text{sq}}$ (i.e., the incidence angle of the LOS projection onto the cross-track plane); this isolates the cross-track ground-range geometry from the along-track squint contribution. θ_{elev} is the satellite-centered elevation-plane angle; it differs from the target-side incidence angle defined above by 2.0° at an 18° off-nadir angle and 7.2° at 47° in the zero-squint plane due to Earth curvature at these orbital altitudes, but it preserves the cross-track ordering of candidate views, so using it in the geometric-quality objective of § 3.2 is ordering-consistent. A critical property is that *all* quality-relevant geometry is a deterministic function of the single decision t_i.

Decision Variables.. A schedule consists of a task selection vector $\mathbf{x} \in \{0,1\}^N$, an ordered sequence $S = (s_1, \ldots, s_m)$ of the m selected tasks, and a vector of observation start times $\mathbf{t} = (t_{s_1}, \ldots, t_{s_m})$. All geometric parameters (θ_i, $\psi_{\mathrm{sq},i}$, R_i) are *derived* from t_i rather than independent decision variables.

3.2. Objective Functions

We formulate a three-objective optimization problem that captures the tension between scheduling throughput and SAR-specific imaging quality. All quality objectives use the *mean* over scheduled tasks, making them independent of the number of scheduled tasks $m = |\mathcal{S}|$ and thereby producing a genuine "quantity versus quality" Pareto trade-off.

Objective 1: Coverage Profit.. The normalized coverage profit (Eq. 1) is defined relative to the greedy baseline

$$f_1^* = \frac{1}{f_1^{\text{G-BL}}} \sum_{i \in \mathcal{S}} p_i \tag{1}$$

where $f_1^{\text{G-BL}}$ is the profit achieved by the profit-only greedy scheduler (G-BL, § 4.2) on the same scenario; normalization by G-BL rather than GA-P-BL is conservative for the MOEA coverage deficit (§ 6.1).

Objective 2: Geometric Image Quality.. The ground pixel area of a SAR image is the product of ground-range resolution $\rho_{\mathrm{rg}} = c/(2B\sin\theta)$ and azimuth resolution $\rho_{\mathrm{az}} = D/(2\cos\psi_{\mathrm{sq}})$. We define a dimensionless geometric quality index:

$$f_2 = \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} \sin\theta_{\mathrm{elev},i} \cdot \cos\psi_{\mathrm{sq},i} \tag{2}$$

Higher f_2 indicates a smaller per-task ground-pixel-area factor (better geometric resolution). The quality indices are unweighted means over the scheduled set, orthogonal to the priority weights used in f_1: they reflect the average imaging geometry of the selected tasks rather than biasing quality toward high-priority targets.

Objective 3: NESZ Radiometric Quality.. Noise-Equivalent Sigma Zero (NESZ) measures the minimum detectable backscatter. For Stripmap SAR, NESZ degrades with the cube of slant range, NESZ $\propto R^3$, after accounting for azimuth compression gain (Curlander and McDonough, 1991). Using the full off-nadir angle ξ for the R^3 factor (which already contains the squint

contribution through $\cos\xi = \cos\theta_{\text{elev}} \cos\psi_{\text{sq}}$, so no separate squint factor is needed), we define the radiometric quality index as:

$$f_3 = \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} \cos^3 \xi_i \tag{3}$$

Higher f_3 indicates a lower per-task NESZ factor (better radiometric sensitivity).

Modeling choices behind Eq. 3.. Equation 3 rests on a 3D geometric approximation that writes the slant range as an elevation-plane range at zero squint over $\cos\psi_{\text{sq}}$, yielding NESZ $\propto 1/(\cos^3\theta_{\text{elev}} \cdot \cos^3\psi_{\text{sq}})$; f_3 serves as a relative ordering index.

Modeling approximation and error.. The plane-Earth model omits Earth curvature and the antenna elevation-pattern roll-off (4%/13% slant-range/R^3 underestimation at the 47° edge) and suppresses the ground-reference $\sin\theta$ factor; the suppression itself is ordering-preserving by construction (§3.1). The comparisons in §5.3 and below depend only on the induced ordering—strongest within windows (mean incidence varies by about 1.5° in S1; §6.5), while across 18°–47° the suppressed factor varies by roughly 2.4×; f_3 is a geometry proxy, not an absolute NESZ measure.

Squint magnitude and R^3 dominance.. In the produced schedules the per-task squint is moderate (mean $|\psi_{\text{sq}}|$ of 16–24°), so the $\cos^3\psi_{\text{sq}}$ factor contributes 10–25% of the f_3 range; NESZ $\propto R^3$ captures the dominant geometric dependence, and platform-constant terms do not affect the ordering.

Physical Trade-Off.. The two quality objectives partition SAR imaging performance by mechanism: f_2 spatial resolution (geometric), f_3 signal-to-noise ratio (radiometric). Their optima conflict: f_3 peaks at the zero-Doppler point ($\psi_{\text{sq}} = 0$) whereas f_2 benefits from moderately higher incidence—radiometric cleanliness against ground pixel area. The squint angle enters both objectives (Eqs. 2–3) through the same $\cos\psi_{\text{sq}}$ factor, so squint is an alignment rather than a trade-off axis. Since θ and ψ_{sq} derive from the same LOS vector (§3.1), the per-task f_2 and f_3 contributions share geometric structure, quantified in §6.4.

3.3. Constraints

Observation-window membership is treated as the domain of each start-time variable rather than as a separate numbered constraint:

$$t_i \in [a_i, b_i - d_i] \quad \forall i \in \mathcal{S}. \tag{4}$$

The remaining four constraints couple task selection, acquisition geometry, temporal feasibility, and resource use. The incidence-angle and squint-angle constraints (C1) are enforced during visibility-window generation; C2–C4 are checked inline by every solver.

C1: Incidence and Squint Angle Constraints.. Each selected acquisition must satisfy the platform incidence-angle range, $\theta^{\min} \leq |\theta_i| \leq \theta^{\max}$, and the squint-angle limit $|\psi_{\text{sq},i}| \leq \psi_{\text{sq}}^{\max}$, where $\psi_{\text{sq}}^{\max}$ is the satellite's maximum achievable squint; we adopt $|\psi_{\text{sq}}| \leq 45°$ as a controlled experimental agility envelope. This bound is an experimental setting, not a claim about any specific platform's demonstrated operating mode or capability. Within C1-feasible visibility geometry the usable envelope is effectively narrower: window generation filters at the 10 s sampling grid, and the θ–ψ_{sq} coupling of the LOS vector (the window-level $|\psi_{\text{sq}}|$ 95th percentile is 37° and the maximum 45°) confines feasible acquisitions; the selected schedules concentrate at mean $|\psi_{\text{sq}}|$ of 16–24° (§6.4). Both are observation-geometry limits applied before scheduling. Modeling scope and its limits are collected in §7.4.

C2: Attitude Maneuver and Non-Overlap.. Between consecutive observations, the satellite must rotate through the angular separation of their LOS vectors. Let $\mathbf{l}_k$ and $\mathbf{l}_{k+1}$ be the LVLH-frame LOS vectors at t_{s_k} and $t_{s_{k+1}}$. The angular gap $\Delta\eta_k = \arccos(\mathbf{l}_k \cdot \mathbf{l}_{k+1}/(|\mathbf{l}_k||\mathbf{l}_{k+1}|))$ must be traversed within the available inter-observation time:

$$t_{s_{k+1}} \geq t_{s_k} + d_{s_k} + \frac{\Delta\eta_k}{\omega_{\max}} + \tau_{\text{settle}} \quad \forall k \in \{1, \ldots, m-1\}. \tag{5}$$

This inequality also enforces non-overlapping task-execution intervals. It is the core coupling in our formulation: changing t_{s_k} shifts $\mathbf{l}_k$, affecting both imaging quality (via θ_k and $\psi_{\text{sq},k}$) and transition feasibility (via $\Delta\eta_k$).

The tightness of C2 scales with task density: as N grows, the fraction of task pairs whose required angular transition $\Delta\eta/\omega_{\max}$ exceeds the available inter-observation time increases, tightening the constraint; C2 is the dominant active constraint in most dense-regime scenarios, except where the C3–C4 resource caps bind (see below).

C3–C4: Energy and Memory Budgets.. $\sum_{i \in \mathcal{S}} e_i \leq E_{\max}$ and $\sum_{i \in \mathcal{S}} m_i \leq M_{\max}$, where $E_{\max} = 10^7$ and $M_{\max} = 10^{11}$ (arbitrary units) and typical per-task consumptions are $e_i = 5 \times 10^4$ and $m_i = 5 \times 10^8$. Both caps therefore translate into the same ceiling of 200 concurrently scheduled tasks. That ceiling is inactive in sparse regimes but binding in a subset of dense scenarios, and its effect on the quality measurements is quantified in § 6.2.

4. Methodology

4.1. Solver Overview

Table 1 summarizes the five configurations compared in this study. We describe the solvers in order of increasing sophistication: from greedy heuristics (G-BL, G-SM) through single-objective metaheuristic (GA-P-BL), to multi-objective evolutionary algorithms (MOEA-2, MOEA-3).

Table 1: Solvers: greedy baselines (G-BL, G-SM), single-objective GA with hot-start (GA-P-BL), and NSGA-III variants with two and three objectives (MOEA-2, MOEA-3)

Label	**Type**	**Objectives**	**Hot-Start**
G-BL	Greedy heuristic	max f_1 (profit-only)	n/a
G-SM	Greedy heuristic	max f_1 + squint minimization	n/a
GA-P-BL	Single-objective GA	max f_1 (G-BL seeded)	std = 0.5
MOEA-2	NSGA-III (2-objective)	$f_1^* + f_2$ (geometric quality)	std = 0.5
MOEA-3	NSGA-III (3-objective)	$f_1^* + f_2 + f_3$ (geo.+NESZ)	std = 0.5

Single-objective solvers (G-BL, G-SM, GA-P-BL) optimize only f_1; their f_2 and f_3 values are computed post hoc on the final schedule for comparison with the multi-objective solvers. The five configurations differ primarily in objective-function configuration rather than algorithmic paradigm. This design is intentional: it isolates the effect of adding physical quality objectives, which is the paper's research question. A comparison across algorithmic families (e.g., NSGA-III vs. MOEA/D (decomposition-based) vs. SMS-EMOA (hypervolume-based)) is left to future work (§ 7.4).

4.2. G-BL: Greedy Baseline

G-BL represents the conventional scheduling approach: it makes no use of SAR imaging physics when ordering or selecting tasks. Tasks are sorted by earliest start time t_{start} and greedily inserted into the schedule at their

earliest feasible time. This earliest-time placement fixes each task's observation geometry implicitly through the window's incidence–squint profile rather than through any quality criterion. If the attitude transition constraint (C2) is violated for a candidate task, its start time is delayed within the window until the constraint is satisfied; if no feasible delay exists, the task is dropped. Since G-BL selects observation times solely for feasibility, its f_2 and f_3 values are incidental by-products of the window geometry. We compute these post hoc on the final schedule to serve as the baseline against which physics-aware solvers are compared.

4.3. G-SM: Greedy Squint-Minimized

G-SM shares G-BL's greedy structure (sort by earliest start time) but modifies the observation time selection: within each visibility window, a task is placed at the discretized time step that minimizes squint angle $\psi_{\text{sq}} = \arcsin(|l_x|/R)$. G-SM minimizes squint angle as a computationally cheap proxy for maximizing f_3 (NESZ radiometric quality); the correlation between squint minimization and f_3 improvement is confirmed by the large effect size (matched-pairs sign statistic $\delta_{\text{pairs}} = +0.99$, § 6.3). This strategy serves as a physics-aware greedy baseline: it answers the question "what quality improvement can a simple squint-minimization heuristic achieve, without any multi-objective search?" The squint-minimizing time selection incurs a profit penalty because the optimal-squint time may lie later in the window, reducing the remaining time for subsequent tasks (constraint C2). As with G-BL, f_2 and f_3 are post hoc outputs.

4.4. GA-P-BL: Single-Objective GA with Hot-Start

GA-P-BL tests whether a genetic algorithm optimizing only f_1 can outperform greedy scheduling by exploiting temporal flexibility. Each chromosome encodes a vector of normalized start times $\boldsymbol{\tau} \in [0,1]^N$, where task i's actual start time is $t_i = t_i^{\text{earliest}} + \tau_i \cdot (t_i^{\text{latest}} - d_i - t_i^{\text{earliest}})$, alongside a binary selection vector $\mathbf{x} \in \{0,1\}^N$; both are decision variables (a $2N$-dimensional chromosome). The population of 100 individuals is initialized from G-BL's solution with Gaussian noise (std = 0.5) added to the full $2N$-dimensional chromosome. The GA runs for 200 generations with simulated binary crossover ($p_c = 0.9$, $\eta_c = 20$) and polynomial mutation ($p_m = 0.1$, $\eta_m = 20$). f_2 and f_3 are computed post hoc on the best-found schedule. After optimization, a never-worse-than-baseline guard compares the best GA schedule with G-BL: if the GA's f_1 falls short, the greedy schedule is returned

instead, guaranteeing $f_1^{\text{GA-P-BL}} \geq f_1^{\text{G-BL}}$ by construction. The guard never triggered in the 200 scenarios: GA-P-BL's own output is numerically identical to G-BL in 199 of 200 scenarios (the single exception improves profit by 13.5%; mean f_2 and f_3 are within 0.001 of G-BL at every scale) (§6.1); the coincidence is an attribute of GA-P-BL's own returned solution, not an effect of the guard. GA-P-BL therefore acts as a control that confirms the greedy profit baseline rather than as an independent solver.

4.5. MOEA-2 and MOEA-3: Multi-Objective Evolutionary Algorithms

MOEA-2 and MOEA-3 extend the GA-P-BL chromosome encoding to multi-objective optimization via the NSGA-III framework (Deb and Jain, 2014). Both use a population of 100 individuals over 200 generations with the same genetic operators as GA-P-BL. Reference points are generated via the Das–Dennis systematic approach (Das and Dennis, 1998), producing $H = \binom{n_{\text{obj}}+p-1}{p}$ structured reference directions, where n_{obj} is the number of objectives and $p = 12$ the divisions per objective dimension ($H = 13$ for MOEA-2, $H = 91$ for MOEA-3). The entire population is hot-started from the G-BL solution perturbed by Gaussian noise (std = 0.5), and an at-least-one-task constraint prevents empty solutions from dominating the Pareto front. As every solver is hot-started from the non-empty G-BL solution (§ 4.2), this constraint is inactive in the reported experiments. MOEA-2 optimizes two objectives: f_1^* (normalized profit) and f_2 (mean geometric quality, Eq. 2). MOEA-3 adds f_3 (mean NESZ radiometric quality, Eq. 3) as a third objective. The comparison between MOEA-2 and MOEA-3 directly tests whether adding the NESZ objective (f_3) produces Pareto-front differentiation beyond what f_2 alone captures, or whether the θ/ψ_{sq} geometric coupling renders the third objective redundant in the two-orbit-horizon, single-satellite setting.

5. Experimental Setup

5.1. Scenario Design

Table 2 defines the four scenario classes.

Table 2: Scenario classes: four density levels (S1–S4) with $N = 20, 100, 300, 500$ targets, 50 scenarios per class

Class	Targets	Look Direction	Description
S1	20	Both	Small, low-density
S2	100	Both	Medium, moderate density
S3	300	Both	Medium-large, transition region
S4	500	Both	Large, high-density

All scenarios share a common satellite platform abstracted as a generic C-band Stripmap SAR with bilateral-looking capability. The platform follows a circular sun-synchronous orbit at 693 km altitude (inclination $\approx 97.8°$, LTAN 06:00, dawn–dusk) and carries a C-band Stripmap instrument with antenna length $D = 12.3$ m and bandwidth $B = 56.5$ MHz. It accesses both left- and right-side incidence angles in the range $[18°, 47°]$ (signed convention: $\theta \in [-47°, -18°] \cup [18°, 47°]$). This generic bilateral configuration is selected to exercise the full incidence–squint trade-off space.

Targets are distributed along the satellite ground track within the accessible observation band (cross-track half-width roughly ± 608 km, a ground central angle of about $\pm 5.5°$, corresponding to the $47°$ incidence limit). Within each class, five scenario variants span the target distribution (uniform, clustered, and mixed) and geometry: variants A–C modulate only the target distribution, while variants D and E additionally modulate the observation geometry, with a class-dependent role: S1-D and S3-D use a single orbital revolution (yielding fewer and narrower windows), S2-E places targets at high latitudes (± 40–$70°$), and S4-D narrows the incidence ceiling from $47°$ to $25°$. The 50 scenarios per class consist of these 5 variants $\times$ 10 random seeds each; the density trend reported in §6.2 is robust to these within-class modulations (e.g., the sparse-regime coverage deficit appears across all S1 variants, and the densest S4 class remains at or near full coverage even for the variants with fewer or tighter windows). The trend also holds with variant A alone (distribution-only modulation, identical geometry definition across classes): the MOEA-2 f_2 gain over G-BL is +37.2% (S1), +3.5% (S2), +0.4% (S3), and +0.3% (S4), reproducing the saturation on the pure density axis; the within-schedule f_2–f_3 correlation follows the same trend on this axis (pooled per-task r over the 10 variant-A schedules per class: -0.80, -0.23, -0.15, -0.31 from S1 to S4, mirroring the dilution trend of §6.4 on this smaller

variant-A-only subset). The four density classes (S1–S4) span sparse reconnaissance ($N = 20$) to dense monitoring ($N = 500$). Other intermediate densities ($N = 50, 400$; classes S5–S6 in the scenario library) are omitted to limit computational cost. To directly constrain the transition midpoint, two additional intermediate classes (S7 at $N = 150$ and S8 at $N = 200$, continuing the numbering, 50 scenarios each, generated under the same generic C-band bilateral configuration) were run for MOEA-2 and the two greedy baselines (G-BL, G-SM) and are used solely in the scale-sensitivity analysis in § 6.2; they do not enter the five-solver pooled comparisons.

Each task is assigned a priority weight p_i drawn uniformly from the integer set $\{1, 2, \ldots, 10\}$, and a fixed observation duration $d_i = 30\,\text{s}$. Visibility windows are computed via orbit propagation over two orbital periods ($\sim$200 min), yielding 1–2 windows per target on average, each typically spanning about a minute (median 1.0–1.2 min across the density classes). Candidate observation times are sampled at a 10 s step within each window, and all reported numbers correspond to a fixed scenario manifest (no manifest revision postdates the results). Windows are filtered by incidence-angle, squint, and minimum-elevation constraints at generation time; the quality indices are then evaluated at each scheduled acquisition time, and C1 feasibility is additionally verified over the full 30 s observation interval (a 0.5 s-resolution scan of all 76,190 windows across the four density classes reports zero violations; the median change of squint over any 30 s within-window sub-interval is 11°, 95th percentile 15°).

5.2. Solver Configuration

All five configurations run on a single CPU core to ensure a uniform computational environment. For orientation, single-core wall-clock per scenario is milliseconds for the greedy heuristics and a median of about 22 s (S1), 66 s (S2), 220 s (S3), and 260 s (S4) for the NSGA-III variants; the study's claims concern solution quality rather than computational efficiency, so these are order-of-magnitude indications rather than a benchmark. G-BL and G-SM are deterministic greedy heuristics with no tunable hyperparameters. GA-P-BL and the MOEA variants use the population size, generation budget, and genetic operators described in §4.4 and §4.5, with the entire MOEA population hot-started from the G-BL solution. The NSGA-III selection operator follows the reference-point-based selection of Deb and Jain (2014).

5.3. Evaluation Metrics

We treat per-objective comparisons paired with Cliff's δ effect sizes as the primary quality indicator, because single-point greedy solvers and multi-point NSGA-III fronts contribute different numbers of points to the objective space. HV is reported as a secondary diagnostic only: with origin-anchored reference boxes it is insensitive to cardinality (dominated points add no volume), but it is also blind to decision-space diversity — as shown in § 6.4, its ranking inverts if the single-point G-SM is included (its HV of 0.2341 exceeds MOEA-3's 0.0893), so HV is not interpreted as a standalone quality judgment. Statistical significance is assessed via the Friedman test for overall differences across the five configurations $\times$ 200 scenarios, followed by pairwise Wilcoxon signed-rank comparisons with Bonferroni correction ($\alpha = 0.05/10 = 0.005$, the family of ten pairwise solver comparisons; the four-variant ablation of § 6.5 applies $\alpha = 0.05/12$ within each reported metric, each family of 12 being the three non-physics variants by four density classes against A, and the S7–S8 ablation comparisons are reported as exploratory and lie outside these families; pairs with zero within-pair difference are dropped from the Wilcoxon signed-rank statistic and reported separately); because G-BL and GA-P-BL coincide in all but one scenario (§4.4), the Friedman statistic is conservative toward fewer distinct rankings and is driven largely by within-scenario dispersion of the two multi-point NSGA-III solvers rather than by baseline-vs-baseline differences, since G-BL, GA-P-BL, and G-SM each contribute a single point per scenario. Per-objective and ablation comparisons are reported with raw p-values alongside the Bonferroni threshold. Effect sizes are measured by Cliff's δ, classified following Romano et al. (2006) ($|\delta| < 0.147$ negligible, < 0.33 small, < 0.474 medium, ≥ 0.474 large). For paired within-scenario solver comparisons we report the matched-pairs sign statistic $\delta_{\text{pairs}} = (W - L)/n$, the matched-pairs restriction of Cliff's δ (counting only within-scenario pairs); the conventional unpaired form $\delta = \big(\#\{x_i > y_j\} - \#\{x_i < y_j\}\big)/(nm)$ gives smaller magnitudes for the same comparisons (e.g. +0.57 versus +0.88 for the sparse MOEA-3/MOEA-2 f_3 comparison), and Romano's thresholds are applied to that conventional form wherever both are quoted. Hypervolume is computed with reference point $(0, 0, 0)$ in the min–max-normalized objective space, where the six min/max bounds (f_1: 0.06–1.16; f_2: 0.11–0.56; f_3: 0.45–0.87) are pooled across all solvers, scenarios, and every persisted non-dominated frontier point (not only knee solutions); per-scenario HV is then averaged within and across classes. This shared normalization framework makes cross-solver

HV values directly comparable as relative rankings: every solver's persisted frontier points, including the two-objective MOEA-2 fronts and the single-point greedy solutions, carry all three coordinates (f_3 is evaluated post hoc for solvers that do not optimize it), so each HV is computed in the same normalized three-objective space, subject to the HV caveat stated above and detailed in § 6.4.

For each MOEA scenario, a single representative solution is selected from the feasible non-dominated front as the knee point: the objectives are min–max normalized across that scenario's front, and the point maximizing the normalized sum ($f_1 + f_2$ for MOEA-2, $f_1 + f_2 + f_3$ for MOEA-3) is reported in Table 3 and the per-objective means. The full front is retained for HV and for the Pareto compression analysis (§ 6.4). We verified the selector sensitivity directly: at $N \geq 100$ the knee and best-f_1 points give class-mean f_2 and f_3 within 0.005, because the dense-regime fronts are compressed (S4 spans only about 0.014 in f_2); at sparse S1 the knee and best-f_1 means differ by about 0.04–0.06 in f_2 and f_3, which is expected and intentional: Table 3 reports the balancing knee rather than either single-objective extreme. The S1 fronts bracket the knee ($f_2 = 0.376$, $f_3 = 0.698$, $f_1^* = 0.85$): the best-f_2 point reaches $f_2 = 0.449$ at $f_1^* = 0.58$, and the best-f_3 point reaches $f_3 = 0.738$ at $f_1^* = 0.68$, so the headline figures are conservative compromise values rather than achievable optima. Quality gains are also class-mean figures: because the objectives are unweighted per-task means, a positive class-mean f_2/f_3 gain does not selectively favor high-priority targets, and per-scenario spread persists even at dense classes (S4 MOEA-3 f_2 10th percentile 0.291 vs. G-BL mean 0.336).

5.4. Experiment Design Overview

This study comprises the following classes of experiments. The first is the main experiment, running all five configurations (G-BL, G-SM, the GA-P-BL control, MOEA-2, MOEA-3) on 50 scenarios per density class (S1–S4), totaling 200 scenarios. Each scenario is solved once per stochastic configuration (the scenario seed governs instance generation; the NSGA-III sampling seed is fixed at the library default while operator-level random draws are not pinned), so within-instance search stochasticity is not replicated; the released processed results pin the exact front realizations analyzed. A replicated-run check quantifies the cost of this design: each NSGA-III solver is rerun five times on the same instance for ten variant-A scenarios per class (400 solves total). At the class-mean level, the quality metrics (f_2, f_3) shift by less

than 0.01 in every class, and f_1^* by less than 0.04 in the worst sparse-class pair — well below the solver-vs-baseline gaps the headline conclusions rest on (e.g. 0.10 in f_2 and 0.20 in f_3 at S1). At the scenario level, 91% of the main-run knee values fall inside the five-repeat envelope of the same instance; the largest deviations occur in sparse scenarios, where small schedules can settle into distinct local optima. Pinning the remaining process-level randomness is left as a reproducibility recommendation. In addition, the agility-envelope sensitivity is probed directly on the sparse $N = 20$ subset under tightened squint caps (25°, 15°; §6.7). The second is the ablation experiment, which successively removes the squint component (variant B), the incidence-angle component (variant C), and all physical objectives (variant D) from the MOEA-3 framework, running each variant on the same six scenario classes (300 scenarios; §6.5). The third comprises robustness control experiments (hot-start dependence, perturbation-sensitivity sweep, random-search calibration, and search-budget control), which verify that the main and ablation findings are not artifacts of initialization or parameter choices (§6.6).

6. Results and Analysis

6.1. Overall Solver Performance

Table 3 reports all five configurations across all four scenario classes. The full-scale view shows why the value of multi-objective optimization cannot be judged from the densest scenario class (S4) alone. At the sparsest density S1, MOEA-2 improves f_2 by +33.7% relative to G-BL on the per-scenario-ratio basis (the pooled-mean ratio gives +33%: class means 0.393 vs. 0.295) while accepting a 14% coverage reduction ($f_1^* = 0.86$, knee-point value; the best-coverage front point gives +12.5%), and MOEA-3 raises f_3 by +40% (0.698 vs. 0.497) at a 15% coverage cost ($f_1^* = 0.85$). At S2 the performance gains diminish, and at S3–S4 MOEA coverage and quality converge to the G-BL reference. Across every scale, GA-P-BL output is numerically identical to G-BL in all but one scenario (mean $f_1^* = 1.00$ at every scale; Sec. 4.4), serving as a control that confirms the greedy coverage baseline rather than an independent solver; normalization by G-BL rather than GA-P-BL (§3.2) is therefore conservative for the MOEA coverage deficit, and G-BL-normalized f_1^* values equal the coverage comparison to the single-objective GA. G-SM occupies a different operating regime, exchanging coverage and f_2 for higher f_3.

Recomputing the sparse-regime comparison above with the ground-reference factor retained enlarges the MOEA-3 gain over G-BL from +40% to +94% (ratio of class means at S1; the class-mean ratio moves from 1.40 to 1.94 because the ground-reference factor penalizes the window-edge placement of G-BL more than the MOEA-3 selections): the proxy reading of f_3 (§3.2) is conservative, and the ordering-based comparisons are unaffected. The proxy also excludes ambiguity-limited NESZ regimes, where the noise floor is set by integrated ambiguity power rather than the direct R^3 scaling; the ordering argument covers the direct-scaling component only.

Table 3: Solver performance across all scenario classes (mean ± standard deviation over 50 scenarios per class). Higher values are preferred for f_1^*, f_2, and f_3.

Class	**Solver**	f_1^*	f_2	f_3
S1 ($N = 20$)	G-BL	1.00 ± 0.00	0.295 ± 0.037	0.497 ± 0.044
	G-SM	0.51 ± 0.16	0.389 ± 0.086	0.722 ± 0.058
	GA-P-BL	1.00 ± 0.02	0.295 ± 0.037	0.498 ± 0.044
	MOEA-2	0.86 ± 0.13	0.393 ± 0.055	0.657 ± 0.040
	MOEA-3	0.85 ± 0.12	0.376 ± 0.050	0.698 ± 0.039
S2 ($N = 100$)	G-BL	1.00 ± 0.00	0.317 ± 0.026	0.561 ± 0.060
	G-SM	0.35 ± 0.09	0.357 ± 0.078	0.735 ± 0.055
	GA-P-BL	1.00 ± 0.00	0.317 ± 0.026	0.561 ± 0.060
	MOEA-2	0.98 ± 0.03	0.329 ± 0.030	0.589 ± 0.044
	MOEA-3	0.98 ± 0.04	0.329 ± 0.030	0.597 ± 0.042
S3 ($N = 300$)	G-BL	1.00 ± 0.00	0.342 ± 0.026	0.616 ± 0.042
	G-SM	0.29 ± 0.04	0.317 ± 0.055	0.756 ± 0.036
	GA-P-BL	1.00 ± 0.00	0.342 ± 0.026	0.616 ± 0.042
	MOEA-2	1.00 ± 0.00	0.343 ± 0.026	0.622 ± 0.040
	MOEA-3	1.00 ± 0.01	0.344 ± 0.026	0.623 ± 0.040
S4 ($N = 500$)	G-BL	1.00 ± 0.00	0.336 ± 0.039	0.675 ± 0.074
	G-SM	0.37 ± 0.12	0.291 ± 0.067	0.786 ± 0.046
	GA-P-BL	1.00 ± 0.00	0.336 ± 0.039	0.675 ± 0.074
	MOEA-2	1.00 ± 0.01	0.337 ± 0.038	0.680 ± 0.073
	MOEA-3	1.00 ± 0.01	0.338 ± 0.038	0.680 ± 0.073

Table 4: Hypervolume across 200 scenarios for the two multi-objective solvers, with G-BL as a single-point reference (single-point solvers G-SM and GA-P-BL are omitted; see the point-count asymmetry discussion in § 6.4).

Solver	HV	SD
MOEA-3	0.0893	0.1053
MOEA-2	0.0668	0.0642
G-BL	0.0234	0.0150

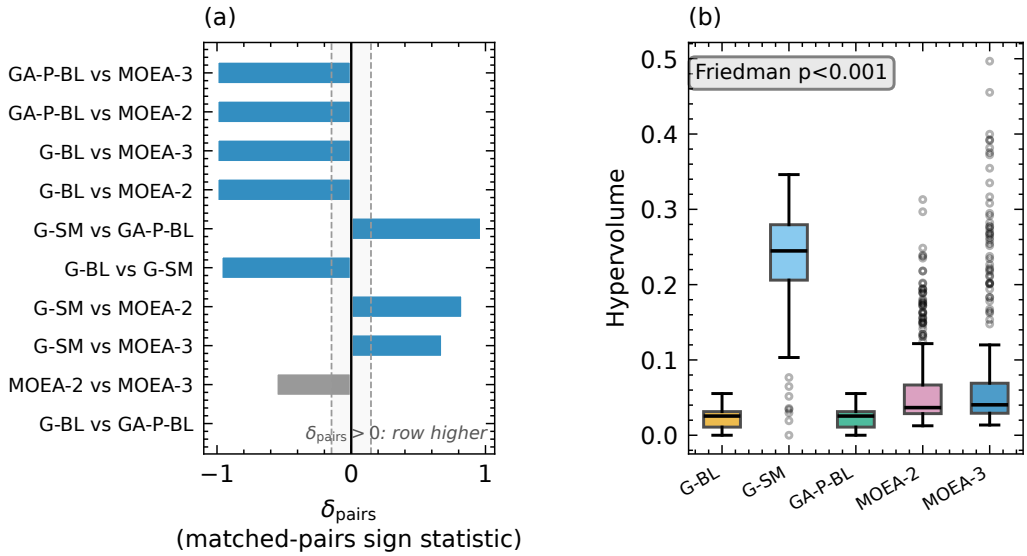


Figure 1: Statistical validation, led by the within-schedule effect-size evidence: (a, left) pairwise effect sizes as the matched-pairs sign statistic $\delta_{\text{pairs}} = (W - L)/n$, the within-scenario analogue of Cliff's δ (the primary evidence); (b, right) hypervolume distribution (secondary diagnostic only, §6.4); the gray band marks the negligible-effect range ($|\delta| < 0.147$ of Romano et al. (2006); shown for reference, as the conventional unpaired δ is weaker for the within-pair comparisons, see §5.3); small grey dots are per-scenario outliers beyond 1.5× the interquartile range (IQR), and the grey bar marks the MOEA-2 vs. MOEA-3 comparison ($\delta = -0.56$).

Fig. 1 summarizes the statistical evidence behind these comparisons. The left panel reports pairwise effect sizes on HV as the matched-pairs sign statistic; bars outside the gray band ($|\delta| < 0.147$) exceed the negligible-effect threshold, and the direction convention is that positive δ favors the row

solver. The right panel shows the per-scenario hypervolume (HV) distribution under a shared min–max normalization; per § 5.3 the HV rankings are secondary diagnostics only, and the point-count confound is analyzed in § 6.4.

The greedy coverage anchor is near-optimal at sparse density: at $N = 20$ it is the exact optimum in 37 of 50 scenarios (32 by the all-targets upper bound, 5 confirmed by an exact CP-SAT model over the 10 s decision grid, released with the analysis scripts); in the remaining 13 the exact optimum exceeds it by a mean of 6.7% (median 5.4%, maximum 14.7% of the exact value), so the anchor carries bounded headroom.

The Friedman test on pooled HV is reported as a descriptive omnibus only: it confirms overall differences among the five solvers across all 200 scenarios ($\chi^2 = 669.02$, $p < 10^{-140}$), and the inferential weight rests on the per-objective pairwise analyses of Table 3. The test setup, the conservative direction of the statistic, and the treatment of zero-difference pairs are specified in § 5.3; interpretation relies on Cliff's δ given the large sample size. MOEA-2 and MOEA-3 differ on pooled HV with $\delta = -0.56$ ($p < 10^{-5}$), but this contrast blends the sparse-regime divergence with the dense-regime convergence and is inflated by the HV point-count asymmetry (§ 6.4); contrasts involving G-SM are large because it occupies an extreme radiometric-quality regime.

The following subsections first map the density landscape—how the multi-objective advantage contracts with target density (§6.2)—then isolate the squint-aware special regime (§6.3), decompose the underlying mechanism (§6.4), test whether the measured gains survive ablation and the initialization and budget controls (§6.5–§6.6), probe the agility-envelope sensitivity (§6.7), and finally quantify the coverage-stratified frontier control against the cheapest placement-only alternative (§6.8).

6.2. Scale Sensitivity

The gap between MOEA-2 and G-BL varies systematically with scenario density (Fig. 2, Table 5). MOEA-2's coverage deficit relative to G-BL is confined to sparse densities: from $f_1^* = 0.86 \pm 0.13$ at S1 ($N = 20$) it rises to 0.98 ± 0.03 at S2 ($N = 100$), and reaches 0.996–0.999 at $N \geq 150$ (S7–S8, S3), remaining within 1% of G-BL at S4 (0.996). The f_2 improvement over G-BL follows the same contraction, falling from +33.7% (S1) to +3.6% (S2) and to +1.3% or less at $N \geq 150$. Scenario-level frequency tells the same story: at S1, 80% of scenarios show a measurable coverage deficit ($f_1^* < 0.95$, i.e., a deficit exceeding 5%), at S2 10%, and 0% at $N \geq 150$. The coverage cost

of multi-objective optimization is therefore a sparse-density phenomenon: between $N = 20$ and $N = 100$ the solver transitions from actively trading off coverage for quality to operating at the greedy baseline, and beyond $N = 150$ the deficit is effectively zero.

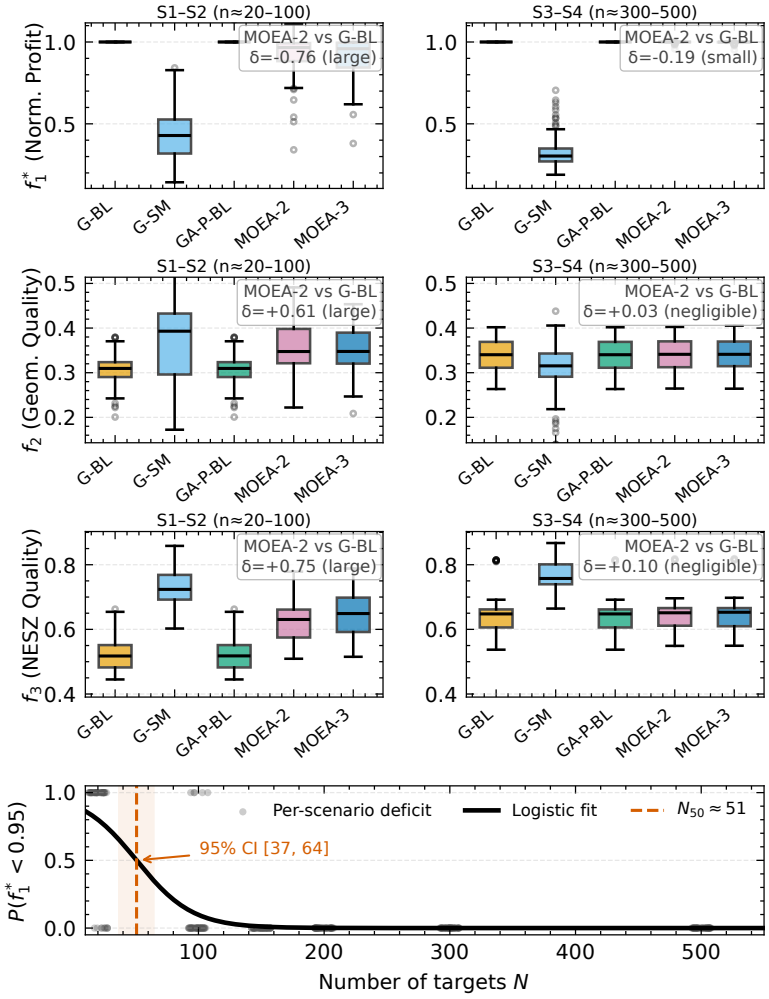


Figure 2: Scale sensitivity: (top) per-objective distributions of normalized profit f_1^*, geometric quality f_2, and radiometric quality f_3 for sparse (S1–S2) and dense (S3–S4) scenario classes; (bottom) logistic fit to the proportion of scenarios with a coverage deficit ($f_1^* < 0.95$), with jittered per-scenario indicators.

Table 5: Scale sensitivity of MOEA-2 vs. G-BL. f_2 improvement is the mean of per-scenario ratios (+33.7% at S1 on this basis; the ratio of pooled means gives +33%). S7 and S8 are intermediate scale points run for MOEA-2 and the two greedy baselines only (§5.4); classes are ordered by target count N.

Class	N	f_1^* (MOEA-2)	f_2 improvement	% $f_1^* < 0.95$
S1	20	0.86 ± 0.13	+33.7% ($\pm$14.3)	80%
S2	100	0.98 ± 0.03	+3.6% ($\pm$3.6)	10%
S7	150	1.00 ± 0.01	+1.3%	0%
S8	200	1.00 ± 0.00	+0.8%	0%
S3	300	1.00 ± 0.00	+0.4% ($\pm$0.2)	0%
S4	500	1.00 ± 0.01	+0.3% ($\pm$0.2)	0%

Fitting a logistic regression to the proportion of scenarios with $f_1^* < 0.95$ across the six density points ($N = 20, 100, 150, 200, 300, 500$) yields a transition midpoint $N_{50} \approx 51$ targets. This estimate is weakly constrained: only the two sparse densities ($N = 20, 100$) carry informative deficit proportions (80% and 10%), while the four denser points are all at most 1%, so the bootstrap confidence interval is wide ($N_{50} \in [37, 64]$). N_{50} should therefore be read as a coarse indicator that the transition is confined to $N \in [20, 100]$, not as a precise threshold. Beyond $N \approx 150$, no scenario shows a coverage deficit exceeding 5%. Refitting the same logistic on the cap-free (C2-limited) subset alone leaves the midpoint unchanged (pooled refit $N_{50} = 50.8$, CI $[38.1, 63.1]$; cap-free subset $N_{50} = 50.8$, CI $[37.5, 62.8]$, $n = 253$), and the 47 ceiling-bound scenarios—all in the dense classes—show no deficit (0/47): the density boundary is a sparse-regime phenomenon, not an artefact of the bound/unbound mix.

The ceiling underlying this split is the C3–C4 resource cap of §3.3: it is inactive in sparse regimes but binds in a subset of dense scenarios—20 of 50 G-BL schedules at S3 and 27 of 50 at S4 select exactly 200 tasks, so the binding constraint there is C3–C4 rather than C2, while the remaining dense scenarios are schedule-limited by C2. Splitting each dense class by this classification, the two subpopulations carry distinct quality profiles: at S4 the C3–C4-bound scenarios average $f_2 = 0.365$, $f_3 = 0.643$ (G-BL class means), whereas the C2-limited scenarios average 0.303 and 0.714 — coverage packing under the shared cap thins per-task quality — while at S3 the two groups are closer (0.346, 0.606 vs. 0.339, 0.623). MOEA-3 reproduces the same split (S4: 0.366/0.647 vs. 0.305/0.720), so class-mean quality margins partly reflect the

bound/unbound mix within each class. The A–D ablation invariance is, if anything, reinforced when all variants are pinned to a shared resource cap (§ 6.5), but dense-regime coverage convergence should be read as convergence to the feasible task count under both constraint families rather than to temporal saturation alone. The f_3 row of Fig. 2 shows the same sparse-to-dense contraction on the radiometric objective: at sparse density MOEA-3 separates visibly from G-BL and G-SM sits in its own high-f_3 regime, whereas the dense panels show all solvers converging to nearly identical distributions, consistent with Table 3. Acquisition timeliness, now measured rather than assumed: relative to the earliest window starts, G-BL itself drifts by class-means of 100–652 s because greedy sequencing defers tasks cumulatively, so the knee-vs-G-BL difference is the meaningful quantity—and it is confined to the sparse regime: the quality-steering knee schedules lag G-BL by 167 s on average at S1, while at S2–S4 the class-mean delays agree within 4 s.

6.3. Squint-Aware Special Regime

The squint-minimized greedy solver (G-SM) improves NESZ radiometric quality (f_3) relative to the profit-only baseline (G-BL) by selecting observation times that minimize squint angle within each visibility window. Across all 200 scenarios, this produces a large positive effect on f_3 (matched-pairs sign statistic $\delta_{\text{pairs}} = +0.99$, $p \approx 1.5 \times 10^{-34}$, on a per-scenario paired basis: 199 of 200 scenarios improve) at an aggregate cost in coverage ($\delta_{\text{pairs}} = -1.0$, $p < 10^{-34}$, on raw f_1; f_1^* is not used here because G-BL yields constant $f_1^* = 1.0$ by definition, for which the normalized-profit comparison degenerates to a one-sample test against a fixed value). The trade-off, however, is strongly density-dependent. In sparse scenarios (S1, $N = 20$), G-SM drops to $f_1^* = 0.51 \pm 0.16$ (-49% vs. G-BL at 1.00) while improving f_3 from 0.497 ± 0.044 to 0.722 ± 0.058 ($+45\%$). In dense scenarios the f_1 cost remains severe: it peaks at S3 ($N = 300$: $f_1^* = 0.29 \pm 0.04$, -71%), eases slightly at S4 ($N = 500$: 0.37 ± 0.12, -63%), while f_3 rises from 0.675 ± 0.074 to 0.786 ± 0.046 ($+16\%$ at S4). Fig. 3 visualizes this density-dependent trade-off: the Locally Estimated Scatterplot Smoothing (LOESS) trend in panel (c) identifies a transition region around $N = 100$ where the f_1 penalty grows rapidly. The radiometric advantage narrows with density ($+45\%$ at S1 to $+16\%$ at S4) as tightly packed time windows constrain how far each acquisition can be moved toward the squint-minimizing time, while the coverage cost remains large at every density (G-SM schedules, on average, 44% of the tasks G-BL selects at S1, falling to $\sim$20–27% at $N \geq 100$; these are means

of the per-scenario scheduled-task-count ratios, the profit-side analogue of which appears in Table 3).

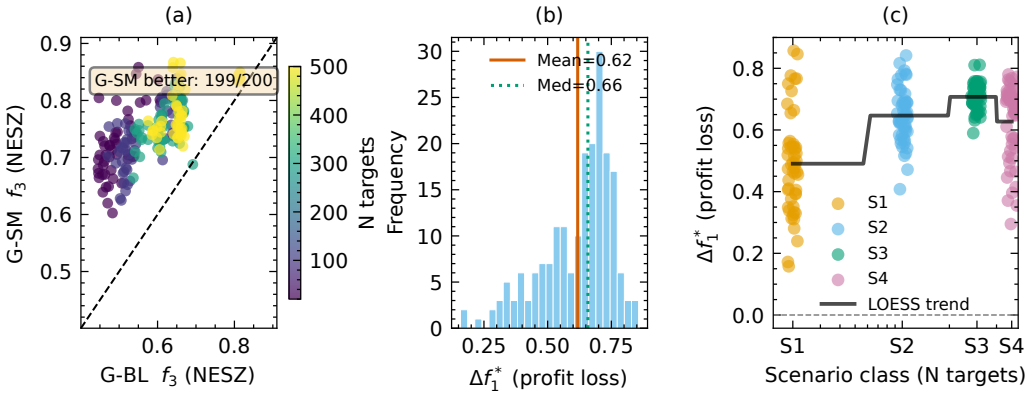


Figure 3: Effect of squint-aware scheduling: (a) f_3 scatter; (b) Δf_1^* histogram; (c) Δf_1^* vs. N with LOESS trend

6.4. Mechanism of Pareto Compression

The dense-regime convergence reported in §6.2, before the special regime of §6.3, has a geometric cause: the two physical objectives conflict most strongly when the scheduler can move acquisitions, and coverage pressure takes that freedom away.

Compression mechanism.. Fig. 4 shows a representative S4 front that contains 4.7 ± 4.4 MOEA-3 non-dominated solutions (MOEA-2: 1.6 ± 0.9) yet spans only about 0.08 in f_1^* and 0.014 in f_2, against a total span of $\Delta f_2 \approx 0.14$ in sparse S1 scenarios. The MOEA-2 and MOEA-3 fronts nearly coincide at this density, so the third objective adds little differentiation there. The compression characterises the G-BL-seeded front near the full-coverage working point, not the feasible surface as a whole: the squint-minimizing heuristic G-SM reaches $f_3 \approx 0.79$ at $f_1^* \approx 0.37$ at S4, a region outside the MOEA-3 surface (§6.3).

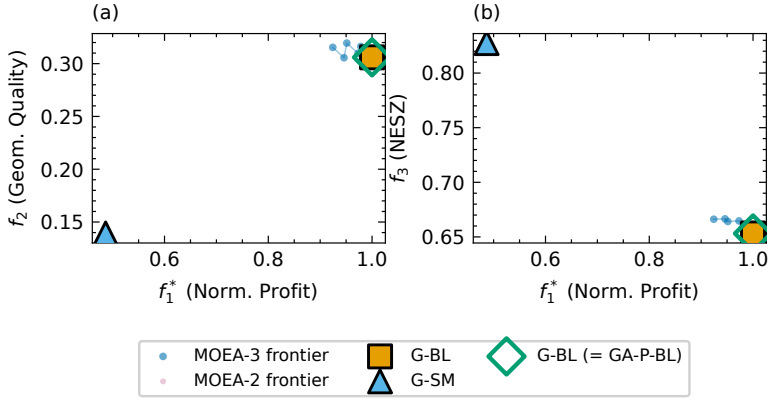


Figure 4: Pareto front on a representative S4 scenario ($N = 500$): MOEA-3 frontier with MOEA-2 overlay and baseline solver markers. Panel (a): f_1^*–f_2; panel (b): f_1^*–f_3.

Null hypothesis and coupling decomposition.. Because θ and ψ_{sq} both derive from the same LOS vector $\mathbf{l} = (l_x, l_y, l_z)$ (§ 3.1), the per-task f_2 and f_3 contributions share geometric structure. We separate this definitional coupling from scheduler-induced correlation with an independent-angle null over the accessible envelope ($\theta_{\text{elev}} \in [16°, 41°]$, $|\psi_{\text{sq}}| \leq 45°$, the experimental squint envelope of §3.3; the satellite-side counterpart of the C1 incidence range $[18°, 47°]$). A permutation null that shuffles the visibility-window-grid θ_{elev} and ψ_{sq} marginals within each density class gives $r_{\text{null}} = -0.60$ to -0.62 across classes (observed shuffle range within ± 0.02 over 20 shuffles per class; a nominal 95% interval would require an order of magnitude more draws), and Monte Carlo sampling over the uniform box gives $r_{\text{MC}} = -0.28$ ($N_{\text{MC}} = 2 \times 10^6$). We adopt the empirical-marginal permutation value as the factorized null, because the permutation preserves the empirical marginals and is therefore the correct null for "same marginals, no LOS coupling", whereas the uniform-box Monte Carlo assumes an idealized geometry:

$$r_{\text{null}} = \frac{\text{Cov}(f_2, f_3)}{\sqrt{\text{Var}(f_2)\,\text{Var}(f_3)}} \approx -0.60, \tag{6}$$

where independence under the null factorizes the moments, e.g. $E[f_2 f_3] = E[\sin\theta_{\text{elev}} \cos^3\theta_{\text{elev}}]\, E[\cos^4\psi_{\text{sq}}]$. The negative sign marks a structural conflict: in the θ_{elev}-only limit (fixed squint), $f_{2,i} \propto \sin\theta_{\text{elev}}$ and $f_{3,i} \propto \cos^3\theta_{\text{elev}}$ are strictly anti-monotonic ($r_\theta = -0.997$ in Monte Carlo), since f_2 favours high incidence and f_3 favours low incidence; in the ψ_{sq}-only limit both rise with

$\cos\psi_{\mathrm{sq}}$ ($r_\psi = +0.997$). The wide-envelope Monte Carlo value (-0.28) is weaker than the broadside limit because the positive ψ-common-mode dilutes the θ_{elev} conflict.

Correlation inside schedules. Fig. 5 reports the pooled per-task correlation within solver-selected schedules for the four objective variants. For every variant the correlated pair is the same fixed full-physics (f_2, f_3) (Eqs. 2 and 3), evaluated on that variant's own schedules; the variant-specific objectives steer task selection but are never themselves the correlated quantities. For the full-physics variant A it is -0.62 at S1 and -0.15, -0.13, -0.23 at S2–S4, with per-schedule means of -0.72 and $-0.15/-0.16/-0.18$ (Fisher-z means $-1.09/-0.16/-0.16/-0.19$ over the 50 schedules per class), and individual S1 schedules span $r \in [-0.97, -0.03]$, all negative. Sparse schedules therefore recover the independent-angle conflict of Eq. 6, while dense schedules weaken below the envelope value. The envelope itself, evaluated over all visibility-window time steps (20 scenarios, five per class, 10 s resolution; 7.6×10^4 geometry samples), gives $\mathrm{corr}(f_2, f_3) \approx -0.36$ (-0.35 at S1, -0.37 at S2–S3): real visibility geometry couples θ_{elev} and ψ_{sq} through the LOS vector, and the resulting positive common mode dilutes the θ_{elev} conflict. Every solver shifts acquisitions toward lower squint than the envelope average (mean $|\psi_{\mathrm{sq}}|$ 16–24°), and coverage packing weakens the correlation further at high density.

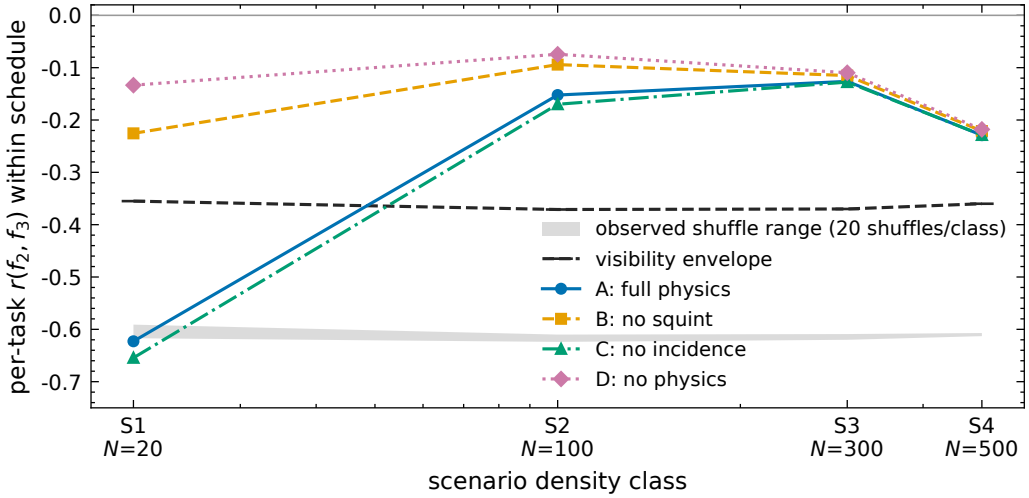


Figure 5: Correlation between the geometric (f_2) and radiometric (f_3) objectives inside solver-selected schedules, as a function of target density, for the four objective variants of §6.5. For every variant the correlated pair is the fixed full-physics f_2/f_3 (Eqs. 2 and 3), evaluated on that variant's schedules; the variant-specific objectives steer selection only. Markers give the pooled Pearson correlation over all scheduled tasks in the 50 scenarios per class; the dashed black line is the same correlation evaluated over all visibility-window time steps (envelope), and the shaded band is the observed shuffle range of the independent-angle shuffle null (20 shuffles per class). Values are computed from the same result sets as Table 3.

Which objective component steers the conflict. Variant B removes squint from the objectives ($f_2 = \sin\theta_{\text{elev}}$, $f_3 = \cos^3\theta_{\text{elev}}$) and the sparse correlation collapses to -0.23, close to the envelope value -0.36; variant C removes incidence ($f_2 = \cos\psi_{\text{sq}}$, $f_3 = \cos^3\psi_{\text{sq}}$) and the correlation holds at -0.65, indistinguishable from variant A; variant D removes the physical objectives altogether and falls to -0.13 at S1 and -0.07 at S2. Configurations that see ψ_{sq} select schedules spanning the θ_{elev} trade-off, whereas configurations blind to squint leave the within-schedule geometry at envelope statistics. The mechanical alternative, that dense correlations weaken because the realized f_2 range contracts, does not hold: the pooled within-schedule standard deviation of the *per-task* f_2 component is 0.16–0.17 at S1–S3 and 0.14 at S4 for variant A, with a pooled per-task range of 0.64–0.65 across classes — a different object from the schedule-level f_2 objective whose pooled bounds (0.11–0.56) enter the HV normalization of § 5.3. The mechanism is alignment: reducing ψ_{sq} simultaneously improves both resolu-

tion components through the same $\cos\psi_{\mathrm{sq}}$ factor (azimuth resolution enters f_2 directly, and $\cos\xi = \cos\theta_{\mathrm{elev}}\cos\psi_{\mathrm{sq}}$ carries the same factor into the R^3 radiometric index), so the residual two-axis structure $[\theta_{\mathrm{elev}}, \psi_{\mathrm{sq}}]$ carries its genuine conflict on the θ_{elev} axis alone (f_2 grows with $\sin\theta_{\mathrm{elev}}$, f_3 with $\cos^3\theta_{\mathrm{elev}}$ at fixed squint, while ψ_{sq} enters both objectives with the same sign). In practice, the zero-squint point coincides with the time of closest approach, which minimizes the slant range and, for a spherical Earth, approximately minimizes the off-nadir angle ξ (i.e. maximizes $\cos\xi$ and hence f_3); solver convergence to low-squint regimes removes the alignment axis and leaves the θ_{elev} trade-off, whose exploitation is what saturates under coverage pressure at high density (the factorized null of Eq. 6 and its axis decomposition above separate the definitional coupling from the scheduler-induced component). All fronts reported here are approximate, obtained by NSGA-III under the tested search budget.

Value of the third objective.. MOEA-3 separates from MOEA-2 in f_3 mainly at sparse density. At S1 it attains 0.698 ± 0.039 against 0.657 ± 0.040 (+6.2% under the reported knee rules, paired Wilcoxon $p = 5.5 \times 10^{-13}$; matched-pairs sign statistic +0.88, 47 of 50 scenarios; conventional Cliff's $\delta = +0.57$) at a small f_2 cost (0.376 ± 0.050 vs. 0.393 ± 0.055, -4%, $p = 0.017$). Because each solver's knee maximises its own objective sum (§ 5.3), we re-selected both fronts under a matched radiometric-aware rule, which gives a controlled margin of +4.9% ($p = 2.3 \times 10^{-11}$, 47 of 50), against +0.05% ($p = 0.86$) for a radiometric-blind rule. The two rules apply the identical procedure (min–max normalisation within each front, then the point maximising the normalised sum; f_3 is evaluated post hoc on every point of both fronts) and differ only in the radiometric term, so the +4.9% is the objective's own contribution and the residual gap of the headline comparison comes from knee selection. The margin survives a change of the reference-direction count: MOEA-3 at S1 with Das–Dennis $p = 4$ ($H = 15$, comparable to MOEA-2's $H = 13$) gives $f_3 = 0.702 \pm 0.041$, statistically indistinguishable from the $p = 12$ value 0.698 ($H = 91$), so the gain does not come from the 7× larger reference-direction set. At S4 the class means coincide (0.680 ± 0.073 for both), although MOEA-3 keeps a consistently signed per-scenario edge (33 of 50, sign statistic +0.32, $p = 4.5 \times 10^{-3}$; conventional Cliff's $\delta = +0.03$, i.e. negligible in distribution). Under the elevation-plane decomposition of §3.2, MOEA-2 already attains high f_3 post hoc, so the marginal value of optimising f_3 explicitly is bounded; at operational densities coverage pressure locks

the selection, while the underlying f_2–f_3 trade-off persists, with a negative within-schedule correlation at every density.

Hypervolume as a secondary diagnostic.. Table 4 ranks G-BL first in coverage and last in HV, and the NSGA-III variants first and second in HV despite their lower coverage. G-SM, omitted from the table, attains the highest HV of any solver (0.2341) from its low-coverage, high-f_3 operating point ($f_1^* \approx 0.37$, $f_3 \approx 0.79$ at S4), so the ranking would invert if it were included. HV rewards an extreme single point and is insensitive to cardinality: every single-objective solver contributes one point per scenario, while MOEA-3 contributes 33.8 ± 8.4 non-dominated points at S1 (MOEA-2: 6.8 ± 2.1). We therefore report HV as a secondary diagnostic only. Consistent with the saturation above, the per-density hypervolume contracts at higher density: at S4 the pooled HV is 0.25 (MOEA-2) and 0.17 (MOEA-3) of its S1 value, reflecting the near-point front compression documented above and not a degradation of solution quality. In one of the 200 scenarios a sparse-density MOEA solution exceeds $f_1^* = 1.00$ (+10.8% above G-BL), so the greedy schedule is a near-optimal coverage anchor (§ 6.1). The front compression of §6.4 does not depend on the HV ranking.

Non-optimality of the discarded set.. Because each quality objective is a mean over the scheduled set, one must ask whether solvers raise it by discarding poorly-geometried targets. They do not: the targets the solvers drop have, if anything, *higher* best-achievable radiometric quality than the retained set (mean window-best $\cos^3 \xi$: 0.92 dropped vs. 0.80 retained for MOEA-3 at S1, 0.95 vs. 0.79 for MOEA-2; gap 0.86 vs. 0.82 at S4). Dropping is not quality-ranked — it prices coverage profit and scheduling feasibility, not per-task quality — so the quality gains reported above come from timing selection within windows, not from cherry-picking the target set.

6.5. Physical-Objective Ablation

To quantify the contribution of incidence-angle (θ) and squint-angle (ψ_{sq}) modeling to solver performance, we repeat the MOEA-3 experiment with four objective-function variants:

A. Full physical (baseline) $f_2 = \frac{1}{|\mathcal{S}|} \sum \sin \theta_{\mathrm{elev},i} \cdot \cos \psi_{\mathrm{sq},i}$, $f_3 = \frac{1}{|\mathcal{S}|} \sum \cos^3 \xi_i$ (identical to the standard configuration in § 6.4).

B. No squint $f_2 = \frac{1}{|\mathcal{S}|}\sum \sin\theta_{\mathrm{elev},i}$, $f_3 = \frac{1}{|\mathcal{S}|}\sum \cos^3\theta_{\mathrm{elev},i}$ (removes ψ_{sq} component; θ_{elev} is the elevation-plane incidence angle, $\cos\theta_{\mathrm{elev}} = \cos\xi/\cos\psi_{\mathrm{sq}}$).

C. No incidence $f_2 = \frac{1}{|\mathcal{S}|}\sum \cos\psi_{\mathrm{sq},i}$, $f_3 = \frac{1}{|\mathcal{S}|}\sum \cos^3\psi_{\mathrm{sq},i}$ (removes θ_{elev} component).

D. No physics $f_2 = f_3 = 1$ (constant per-task objective; solver receives no geometric information; f_2, f_3 shown in Table 6 are post hoc diagnostics computed from the actual observation geometry of the resulting schedule). Variant D serves as a null-model control: it tests whether the NSGA-III solver's performance is attributable to the physical content of f_2 and f_3, or merely to the presence of additional objectives that structure the search.

All four variants use identical NSGA-III parameters (population = 100, generations = 200, Das-Dennis $p = 12$), the same scenario files, and the same G-BL hot-start injection. Only the objective functions differ, making this a controlled ablation rather than a retuning experiment. Within-solver ablation with identical NSGA-III parameters and the same G-BL seed injection controls confounding factors sufficiently that the Wilcoxon signed-rank test is appropriate for paired comparison of objective-function variants. Each variant completes six scenario classes × 50 seeds = 300 runs (total: 1,200 runs across A–D); Table 6 reports the density-regime classes S1–S4. The two remaining classes stress non-density axes: S5 holds density at $N = 20$ but spreads target positions along-track over 5°/15°/25°/35°/45° (a squint-spread sensitivity group), and S6 packs the 20 targets into 2–10 clusters with narrow windows (60–180 s; a C2 transition-stress group). Degradation is reported as $\Delta\% = (\mathrm{A} - \mathrm{V})/\mathrm{A} \times 100$, computed on *raw profit* means (it is therefore not exactly reproducible from the G-BL-normalized f_1^* column), where A is the full physical baseline and V is the variant; negative values indicate the variant outperforms the baseline on f_1^*. Percentages quoted for the quality metrics f_2 and f_3 in the text follow the direct relative-change convention $(\mathrm{V} - \mathrm{A})/\mathrm{A}$ (negative values indicate degradation), the opposite sign of the $\Delta f_1^*\%$ column; the bars of Fig. 6 for all three metrics follow the $\Delta\% = (\mathrm{A} - \mathrm{V})/\mathrm{A}$ convention, so a positive bar always means the variant is worse than A.

Table 6: Per-class ablation results. f_1/n is the mean priority weight of the selected tasks. Bold: significant at the ablation threshold $\alpha = 0.05/12$ on the $\Delta f_1^*\%$ column (Wilcoxon signed-rank vs. A, §5.3; for this metric the family comprises the 12 variant-by-class comparisons, and all marked cells satisfy it).

Class	**V**	f_1^*	f_2	f_3	f_1/n	$n_{\mathbf{sel}}$	$\Delta f_1^*\%$
S1 ($N = 20$)	A	0.850 ± 0.123	0.376	0.698	5.99 ± 0.73	15.6 ± 2.6	(baseline)
	B	0.852 ± 0.152	0.353	0.574	6.01 ± 0.83	15.7 ± 3.1	-0.8%
	C	0.845 ± 0.095	0.375	0.703	5.86 ± 0.80	16.0 ± 2.2	$+0.1\%$
	D	1.002 ± 0.011	0.302	0.525	5.65 ± 0.66	19.5 ± 0.7	$\mathbf{-18.0}\%$
S2 ($N = 100$)	A	0.976 ± 0.040	0.329	0.597	5.58 ± 0.42	62.7 ± 19.3	(baseline)
	B	0.990 ± 0.022	0.321	0.570	5.55 ± 0.37	63.6 ± 19.1	$\mathbf{-1.2}\%$
	C	0.977 ± 0.033	0.328	0.599	5.57 ± 0.40	62.8 ± 19.3	$+0.0\%$
	D	1.000 ± 0.000	0.317	0.562	5.54 ± 0.37	64.2 ± 18.8	$\mathbf{-2.0}\%$
S3 ($N = 300$)	A	0.998 ± 0.005	0.344	0.623	5.63 ± 0.30	154.2 ± 39.9	(baseline)
	B	0.998 ± 0.005	0.343	0.618	5.63 ± 0.29	154.2 ± 40.0	$+0.0\%$
	C	0.998 ± 0.004	0.344	0.623	5.63 ± 0.30	154.2 ± 39.9	$+0.0\%$
	D	1.000 ± 0.000	0.342	0.616	5.63 ± 0.29	154.5 ± 39.9	$\mathbf{-0.2}\%$
S4 ($N = 500$)	A	0.997 ± 0.008	0.338	0.680	5.83 ± 0.44	142.3 ± 72.1	(baseline)
	B	0.999 ± 0.003	0.337	0.677	5.84 ± 0.44	142.5 ± 72.0	-0.1%
	C	0.998 ± 0.006	0.337	0.680	5.83 ± 0.44	142.4 ± 72.0	$+0.0\%$
	D	1.000 ± 0.000	0.336	0.675	5.83 ± 0.45	142.7 ± 71.9	$\mathbf{-0.2}\%$

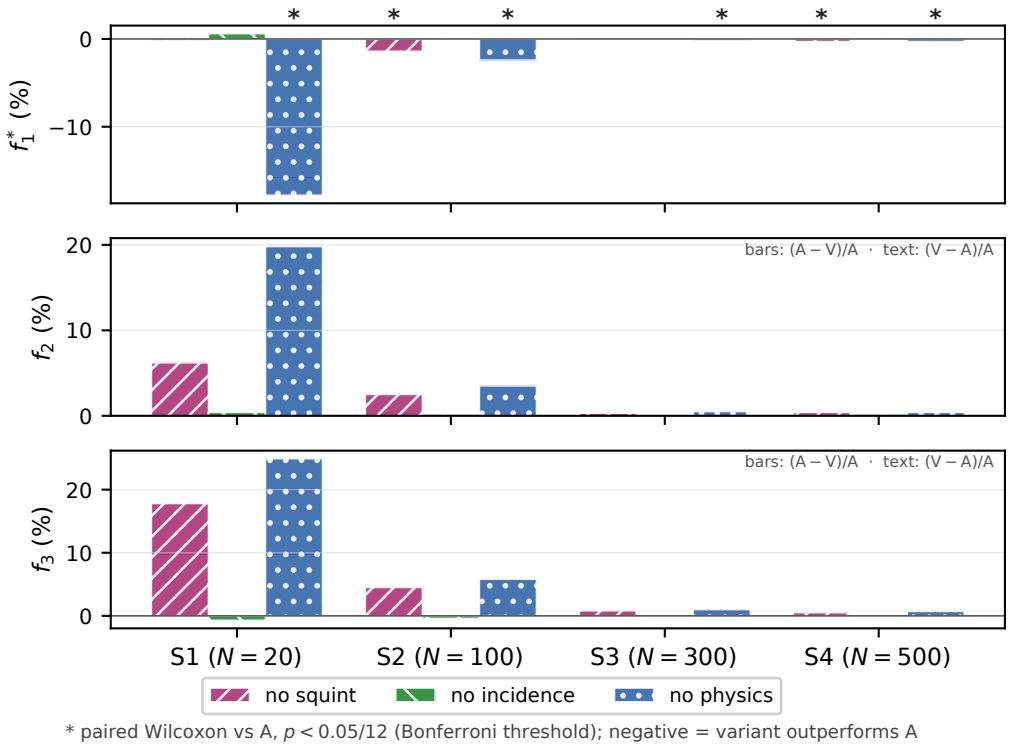


Figure 6: Ablation results across objective-function variants B–D and scenario classes S1–S4. Bars show degradation relative to the full-physics baseline A for normalized profit f_1^* and post hoc geometric-quality metrics f_2 and f_3; negative values indicate that a variant outperforms A. Asterisks mark the paired Wilcoxon comparisons of f_1^* against A significant at the Bonferroni threshold ($p < 0.05/12$); the corresponding per-metric families for f_2 and f_3 are reported in the text and not marked in the figure.

Table 6 and Fig. 6 probe how far the sparse-regime mechanism of § 6.2 survives when components of the physical model are removed.

At S1 ($N = 20$), only variant D (no physics) differs significantly from baseline A at the ablation threshold ($\alpha = 0.05/12$): its f_1^* rises to 1.002 (-18.0%, $p = 3.4 \times 10^{-9}$) by scheduling 3.9 more tasks (19.5 vs. 15.6) at a substantial radiometric cost, with post hoc f_3 falling from 0.698 to 0.525 (-25%). Variant B (no squint) leaves f_1^* unchanged (-0.8%, $p = 0.46$) but lowers f_3 to 0.574 (-18%) and f_2 to 0.353 (-6.2%, $p = 1.1 \times 10^{-5}$): without squint penalization the solver tolerates higher-squint views, sacrificing the radiometric quality that the full-physics baseline purchases in the sparse regime. Variant C (no incidence) produces no measurable change in any re-

ported metric. This inertness is consistent with the window geometry: the elevation-plane incidence angle varies by only a few degrees across a visibility window (mean $\sim 1.5^\circ$ in the S1 scenarios), whereas the squint angle sweeps ~ 20–35°, so the θ_{elev} factor in f_2/f_3 acts as an approximately common scale within each window; removing it preserves the within-window task ordering and hence the selected schedule. The incidence conflict of §6.4 therefore operates through the across-window/aggregate geometry rather than through per-window selection. Thus the physical content of the objectives acts on sparse-regime outcomes chiefly through the squint channel into radiometric quality (f_3; the variant-B-versus-variant-C contrast) and through the full-removal coverage contrast, while partially removing either geometric component leaves coverage unaffected.

A diagnostic arm that replaces the third objective with an exact copy of f_1 (f_2 unchanged) leaves the knee-point deficit in place ($f_1^* = 0.829 \pm 0.150$; Wilcoxon $p = 0.62$ against variant A's 0.850 and $p = 0.32$ against two-objective MOEA-2's 0.859—no detectable difference in either comparison), while the front maximum reaches full coverage in every arm ($\max f_1^* \approx 1.00$; variant D's single-point front makes its knee coincide with that maximum). The coverage cost of variant A is therefore the knee-point position under a real quality trade-off, not a three-objective selection-pressure artefact (S1, 50 scenarios, single-run protocol; variant D itself shows no deficit, which gives the comparison its discriminating power).

At S2 ($N = 100$), variants B and D remain significantly different from A but the effects are small (-1.2%, $p = 5.3 \times 10^{-4}$ and -2.0%, $p = 1.7 \times 10^{-6}$): removing squint or all physics still changes coverage modestly. Variant C remains indistinguishable from A.

At S3–S4 ($N \geq 300$), every variant lies within 0.2% of the baseline for f_1^* and within 0.6% for f_2; D's paired-test significance reflects a consistent but negligible effect (-0.2%). This numerical invariance is the direct signature of the scale-dependent saturation discussed in § 6.2: once the density of candidate observation windows saturates the schedule, the objective function only penalizes infeasible solutions rather than differentiating among feasible ones. In this regime, the NSGA-III solver acts as a constraint-filtered profit maximizer regardless of whether f_2 and f_3 carry geometric meaning (A) or serve as constant placeholders (D).

The two robustness classes replicate this pattern away from the density axis. On S5 (along-track target spread) and S6 (narrow-window clusters), the full-removal variant D is significantly worse than A in f_2 (-11.2%, $p =$

1.6×10^{-13} and -11.1%, $p = 1.8 \times 10^{-15}$) and in f_3 (-6.1% and -6.3%), C is indistinguishable from A (f_2 $+0.4\%$, $p = 0.33$ and -0.6%, $p = 0.48$), and B shows a small f_2 difference that is significant at S6 ($+4.6\%$, $p = 1.2 \times 10^{-5}$) but not at the corrected threshold at S5 ($+3.2\%$, $p = 0.046$). The coverage price of physics persists here as well: D attains 17% (S5) and 8% (S6) more raw profit than A, i.e. objective-driven selection sacrifices coverage in sparse configurations regardless of whether the stressor is density, target spread, or window clustering.

Taken together, the physical quality objectives add value mainly in the sparse regime, and primarily through radiometric quality: at S1 the full-physics A attains 33% higher f_3 than the no-physics D (0.698 vs. 0.525; D lies 25% below A on the common-degradation basis) at an 18% coverage disadvantage, and removing only the squint component (B) already sacrifices 18% of f_3 while preserving f_1^*. The coverage effect of partially removing the objectives is minimal in every regime, and by $N \geq 300$ the choice of physical objectives is numerically irrelevant to outcomes.

6.6. Robustness and Calibration

Fig. 7 collects four control experiments that separate the mechanism above from initialization, perturbation, search-budget, and chance effects.

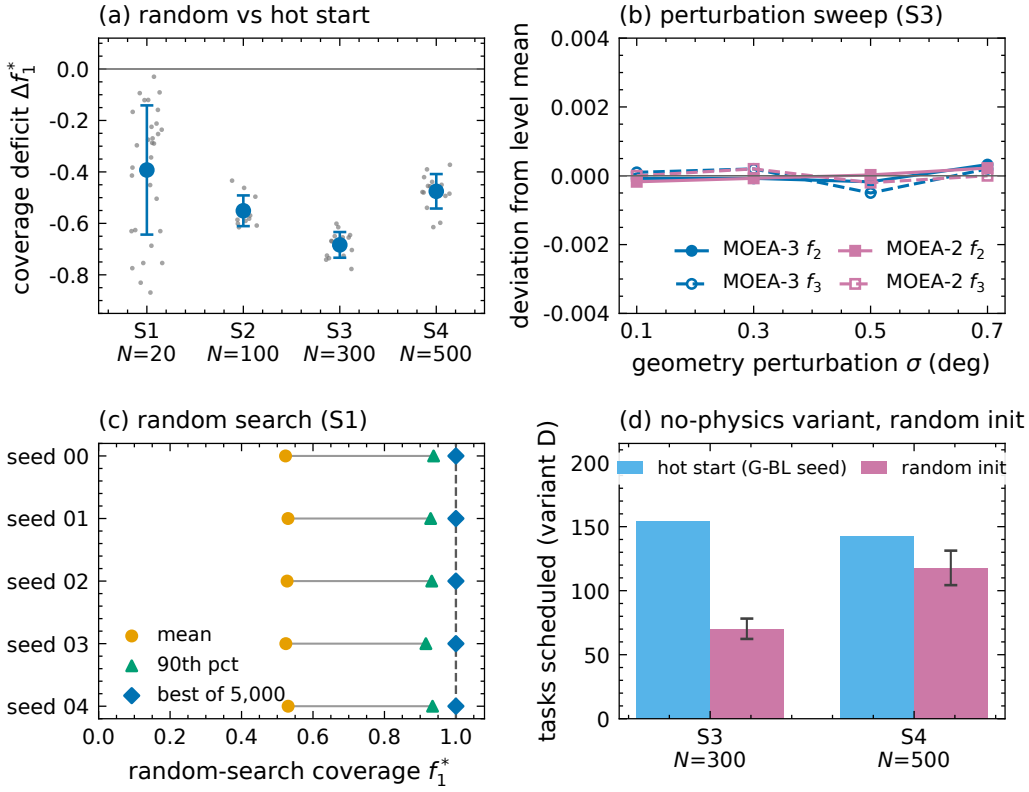


Figure 7: Control experiments of §6.6. (a) Per-scenario coverage deficit of random-initialized relative to hot-started MOEA-2, with mean and SD over the tested scenarios ($n = 30, 12, 15, 15$ for S1–S4; grey dots are individual scenarios). (b) Deviation of MOEA-2 and MOEA-3 quality from its level mean across the geometry-perturbation sweep on ten S3 scenarios per level; error bars omitted because the level-to-level spread is below the plotted range. (c) Coverage reached by 5,000 random task selections on five S1 scenarios, summarised per scenario as mean, 90th percentile, and best sample; the dashed line marks the greedy baseline ($f_1^* = 1$). (d) Tasks scheduled by the no-physics variant D under greedy hot-start (class mean over the 50 scenarios) and random initialization (mean and SD over 15 runs per class).

Initialization.. Random-initialized MOEA-2 loses 0.393 ± 0.251, 0.551 ± 0.060, 0.684 ± 0.050, and 0.475 ± 0.067 in f_1^* at S1–S4 relative to hot-start MOEA-2 (Fig. 7a; 10/4/5/5 scenarios with three seeds each). G-BL seeding therefore drives the coverage convergence. The effect is not confined to coverage: random initialization attains higher f_2 at sparse density (0.505 vs. 0.395 at S1, +28%; 0.460 vs. 0.320 at S2, +44%), marginally higher at S3 (0.349 vs. 0.339, +3%), and lower only at the densest S4 (0.347 vs. 0.380, −9%).

The G-BL anchor's material benefit is coverage, and its geometric advantage is confined to the densest regime. Because the headline experiments hot-start from G-BL, the sparse-regime margins are anchored-front values, and a merged hot-start-plus-random-init front could extend the quality–coverage envelope beyond either run alone.

Perturbation.. A sweep over $\sigma \in \{0.1, 0.3, 0.5, 0.7\}$ on MOEA-3 (10 S3 scenarios per level, fixed solver seed) leaves quality flat: mean $f_2 \approx 0.343$ and mean $f_3 \approx 0.610$ at every level, with level-to-level variation of at most $\sim 1 \times 10^{-3}$ (Fig. 7b; MOEA-2 behaves the same, at $f_2 \approx 0.343$ and $f_3 \approx 0.609$), and coverage stays flat at 200 of the 300 targets (all ten scenarios at the ceiling for $\sigma \leq 0.5$, mean 199.5 at $\sigma = 0.7$; § 3.3). The coupling reported in § 6.4 is a property of the schedule geometry, not of the perturbation magnitude. All control runs use the same Das–Dennis partition ($p = 12$, giving $H = 91$ reference directions for three objectives and $H = 13$ for two) as the headline experiments.

Chance.. On five S1 scenarios, 5,000 random task selections per scenario produce mean $f_1^* = 0.527 \pm 0.003$, a 90th percentile of 0.930 ± 0.008, and a best sample of 1.000, the G-BL profit (Fig. 7c). Unconstrained random selection never exceeded the greedy baseline on these scenarios, an empirical bound on the tested batch: the sampler does not enforce C2–C4, so high coverage arises only by chance.

Search budget and random initialization.. Doubling the budget (population 200, 400 generations) on five S3 and five S4 scenarios keeps the A–D difference in f_1^* negligible (-0.12% at S3, -0.44% at S4; both variants reach full coverage), and shifts quality by at most $+1.64\%$ relative to the standard budget. That shift is the same order as the sub-2% margins it is meant to defend, so the control alone cannot separate premature convergence from all runs reaching the same coverage wall; the bound/unbound split of § 3.3 separates the two subpopulations directly. A doubled-budget control on the five S1 scenarios—the sparse regime where the margins are largest—leaves the A–D difference in place (variant A f_1^*: 0.931 standard vs. 0.924 doubled on the same scenarios; variant D 1.000 in both), so the sparse-regime gap is not an artifact of the standard budget. A random-init run of the no-physics variant D (15 S3 and 15 S4 runs) reproduces the low per-task squint of the hot-started runs ($f_2 = 0.349 \pm 0.016$ at S3 and 0.347 ± 0.015 at S4, against 0.342 ± 0.026 and 0.336 ± 0.039) at a coverage cost: it schedules 70.3 and

117.8 tasks at S3 and S4, against 154.5 and 142.7 for the hot-started variant (50 scenarios per class; Fig. 7d). The lower-squint convergence is therefore not a hot-start residue; if anything the seed biases hot-started runs toward the higher squint of the greedy schedule, which inserts tasks at their earliest feasible time.

6.7. Agility-Envelope Sensitivity

The agility-envelope sensitivity, previously an open qualification, is tested directly: repeating the sparse $N = 20$ configurations with the window-generation squint cap reduced from 45° to 25° and 15° (50 paired scenarios per cap, identical targets and seeds) shows that the multi-objective edge contracts monotonically with the envelope. Relative to G-BL in the same scenarios, MOEA-3's f_2 advantage shrinks from +28% at the 45° cap to +9% at 25° and +2.3% at 15°, while its coverage cost shrinks from 15% to 4% to $\approx 0\%$ ($f_1^* = 0.85$, 0.96, 1.01; the value above 1.00 reflects the near-optimal rather than strict coverage anchor, §6.4); absolute geometry improves for every solver under the tighter cap (G-BL f_2 rises from 0.295 to 0.405 to 0.448), because window filtering removes the low-quality high-squint options.

6.8. Coverage-Stratified Frontier Control

Beyond the coverage–quality exchange rate, MOEA-3's remaining value is the coverage-stratified frontier itself, measured against the cheapest alternative: a placement-only interpolation family (a q-fraction of the greedy coverage schedule re-timed by the squint-minimising rule, $q \in \{0, 0.25, 0.5, 0.75, 1\}$, $K = 10$ random subsets per level, feasibility restored by the same C2 repair the heuristic uses); that family is feasible throughout, and at matched coverage (mixture coverage at or above the knee point's) the knee point dominates the best feasible mixture point in 185 of the 200 scenarios (92.5%) for f_2 and 184 (92.0%) for f_3.

7. Discussion

7.1. Summary of Findings

At $N = 20$ the three dimensions form a genuine trade-off surface. G-BL retains $f_1^* = 1.00$ at a moderate f_3 (0.497), MOEA-2 reaches the highest f_2 (0.393) alongside a f_3 of 0.657, and MOEA-3 trades f_2 (-4% vs. MOEA-2) for the highest f_3 among the NSGA-III configurations (0.698, +6.2% vs. MOEA-2 under the reported knee rules), so no solver dominates on more than

one dimension. At $N \geq 300$ all NSGA-III configurations converge to the G-BL anchor and the tested multi-objective variants provide little additional benefit over the greedy baseline. Our experiment across 200 systematically varied scenarios yields five interconnected findings: **(1)** the quality advantage of multi-objective optimization saturates with target density (+33.7% in f_2 at $N = 20$ falling to +0.3% at $N = 500$); **(2)** the third objective is density-dependent rather than redundant: per-task f_2–f_3 correlation within schedules stays negative at every density, strongest at sparse density (§ 6.4), and MOEA-3 separates from MOEA-2 in f_3 at sparse density (+6.2% at $N = 20$) with the advantage shrinking to $\lesssim 0.2\%$ at $N \geq 300$, confirming the coverage-pressure mechanism detailed in § 6.4. **(3)** G-SM occupies a distinct high-radiometric-quality regime at density-dependent coverage cost; **(4)** solver outcomes become invariant to the choice of physical objectives at $N \geq 300$ (ablation study); and **(5)** coverage convergence toward the G-BL anchor is conditional on the hot start, not a structural property. The first four reflect geometric properties of the two-orbit-horizon observation geometry (finding (2) being an empirical convergence effect under those properties, § 6.4); the fifth reflects solver initialization.

Taken together, these results establish that the G-BL-seeded front we obtain near the full-coverage working point, while offering multiple quality-coverage configurations, is compressed by the geometric coupling above: the narrow f_2 range limits practical differentiation between solutions.

The pooled-HV Friedman omnibus and pairwise contrasts are reported in § 6.1.

Independently of the multi-objective question, GA-P-BL is never worse than G-BL by construction and coincides with it in all but one scenario (§4.4); the present claims therefore concern the marginal value of adding physical objectives rather than profit-oriented search alone.

7.2. Operational Implications

The scale dependence has direct consequences for mission planning. In sparse regimes ($N \leq 100$), MOEA-2 achieves a +33.7% geometric-quality improvement at S1 at a 14% coverage cost ($f_1^* = 0.86$ at $N = 20$ relative to G-BL). Whether this trade-off is acceptable depends on the application: for maritime surveillance, the knee-point gain at S1 corresponds to roughly a 25% smaller per-task geometric area factor (the ground pixel area of a task scales with the reciprocal of the per-task index factor of Eq. 2, evaluated at the elevation-plane angle; equivalently, a linear ground sample dis-

tance about 13% finer at the knee-point geometry ($\sqrt{0.75} \approx 0.87$); the exact schedule-level area change additionally depends on the selected task mix), and the coverage cost may be justified; for area-monitoring missions, the greedy baseline remains preferable. In dense regimes ($N \geq 300$), the multi-objective advantage falls below 1% improvement in f_2, while G-SM offers the highest radiometric quality (+16% f_3 at S4) at a 63% coverage cost; this gain comes from the one-line squint-minimization heuristic rather than from multi-objective search. Read as an exchange rate, this favors the quality-first heuristic when coverage can be spent: G-SM matches or exceeds MOEA-3's radiometric quality at every density (f_3 higher by +3.4%, +23.1%, +21.3%, +15.6% at S1–S4, from the class means of Table 3) while spending 49–71% of the achievable coverage ($f_1^* = 0.51$, 0.35, 0.29, 0.37 against the G-BL anchor of 1.00).

The operational choice is therefore application-side—coverage budget versus achievable quality—rather than solver-side: the coverage-stratified-frontier control (§6.8) shows the knee point dominating the cheapest placement-only alternative. The interpolation family (§6.8) re-times G-BL's selection only—it does not mix task selections, and its $q = 1$ endpoint is not G-SM itself—so the measured comparison bounds the re-timing channel, not selection diversity.

The quality advantage of physics-aware multi-objective scheduling is therefore specific to wide-envelope platforms (§6.7); on platforms with $|\psi_{\mathrm{sq}}| \lesssim 25^\circ$ agility, or without a dedicated large-squint timing/processing chain (§ 3.3), the greedy baseline attains comparable quality at full coverage even in sparse regimes. The logistic midpoint $N_{50} \approx 51$ (§ 6.2) provides a coarse indicator between the density regimes. Our results suggest a conditional operational guideline: for sparse observation requests ($N \lesssim 100$) on squint-agile platforms with the dedicated large-squint timing/processing chain of §3.3 (the 45° experimental envelope of §3.3), MOEA-2 may provide geometric quality gains that outweigh the coverage penalty in specific contexts: the knee point trades 14% coverage for +33.7% f_2, while the best-coverage front point still gives +12.5% f_2 at zero coverage cost, a configuration preferable when coverage cannot be sacrificed; for sparse requests on narrow-envelope platforms, and for dense monitoring campaigns ($N \gtrsim 300$) on the tested platform family, G-BL attains the coverage reference, with the multi-objective variants adding little over it, while G-SM attains the highest radiometric quality at a substantial coverage cost, so the greedy choice follows the mission's priority. Indeed, the MOEA-2 coverage

gap relative to the coverage-only anchor is within 0.4% at $N \geq 300$ (Table 5); since GA-P-BL coincides with G-BL in all but one scenario, no meaningful separate gap exists. The measured timeliness (§6.2) adds a cost to the quality gain only where that gain is on offer, and does not differentiate the dense-regime choice.

7.3. Transferability Across Remote-Sensing Domains

The analytic device most likely to transfer beyond SAR is the envelope–null–schedule decomposition of §6.4: separating the geometric correlation over the full feasible envelope from the within-schedule residual isolates where an objective's value is lost to saturation rather than to a mathematical redundancy. The geometric-resolution objective (f_2) carries directly to optical systems, where ground sample distance grows with off-nadir angle (the cross-track optical ground sample distance scales as $1/\cos^2\eta$, the standard flat-earth off-nadir scaling for a fixed-IFOV pushbroom sensor; the pixel area additionally takes a $1/\cos\eta$ along-track factor), and an analogous density-dependent saturation is a testable expectation for such settings; the radiometric objective (f_3, NESZ) does not, because optical radiometry depends on illumination and cloud cover rather than on observation geometry alone. The decomposition, rather than any single solver ranking, is the transferable contribution.

7.4. Limitations

Our study is bounded by scope decisions that define directions for future work.

Scope.. The density-dependent third-objective indistinguishability reported above is specific to the single-satellite, two-orbit-horizon Stripmap setting: it reflects solver convergence to lower-squint geometries that attenuate the envelope-level f_2–f_3 correlation while the residual within-schedule θ-conflict persists (negative at all densities, strongest at sparse density; §6.4, Eq. 6), not a mathematical identity between f_2 and f_3. In multi-pass or multi-satellite settings with greater angular diversity, the θ-conflict re-emerges and the third objective regains discriminating power; dynamic arrivals and rolling-horizon re-optimization are out of scope here, and their effect on the saturation threshold is untested. ScanSAR, Spotlight, and TOPS modes introduce different NESZ and resolution dependencies beyond Stripmap, and the scenario

design uses a fixed orbit (693 km circular LEO) without exploring sensitivity to orbital parameters. The two-orbit scheduling horizon is likewise fixed: longer horizons grant more scheduling freedom per target, so the density boundary should not be read as horizon-independent (the single-revolution scenario variants are used for robustness checks only, not for the boundary). The quality objectives capture first-order geometric and radiometric effects (ground-range resolution, azimuth broadening, R^3 NESZ scaling). Throughout, ψ_{sq} is the geometric line-of-sight squint, the angle between the target line of sight and the zero-Doppler plane; this is the quantity that enters f_2, f_3, and the C1 squint bound. Second-order radar effects are treated as platform-level constants and are not modeled: the geometric Doppler centroid $f_{\mathrm{dc}} \approx (2v/\lambda)\sin\psi_{\mathrm{sq}}$ reaches 10^2 kHz at the scheduled 16–24° squints, one to two orders of magnitude above typical Stripmap PRFs (10^0–10^1 kHz), so unmanaged echoes would alias outside the processing band and leave the Doppler centroid ambiguous. Yaw steering cannot remove this offset: it compensates only the small Earth-rotation component, whereas the geometric squint is fixed by the target's direction relative to the velocity vector at the scheduled 16–24°. A squinted acquisition therefore requires a dedicated receive-window and PRF timing design, which is outside the model; the projection loss itself is not absorbed by Doppler processing — keeping the beam on the target fixes ψ_{sq}, and the $1/\cos\psi_{\mathrm{sq}}$ broadening in the azimuth resolution is what f_2 charges for. The echo-processing chain itself (receive window, duty cycle, ambiguity control, PRF selection), the azimuth antenna gain rolloff with squint, and squint-dependent azimuth processing are all outside the model; the Doppler magnitude is used only as a scale, whose sign follows the look direction and is out of scope. This idealization preserves the relative ordering of acquisition geometries *within* the simulated platform family, which is what the solver comparisons act on; absolute performance claims for platforms without a dedicated large-squint timing/processing chain are out of scope. The sensitivity of the recommendations to the agility envelope itself is tested directly in §6.7.

Two further scope notes bound the operational reading: the C3–C4 resource caps equate to a 200-task ceiling (§ 6.2) and bind in roughly half of the dense S3–S4 scenarios, so dense-regime convergence reflects both temporal (C2) and resource (C3–C4) saturation, and results are conditional on the tested caps; the dwell time is fixed at 30 s, removing the dwell-versus-SNR trade-off from the model. The utility model optimizes quality and quantity without an explicit latency or urgency objective; acquisition timeliness is not

optimized, but the acquisition-delay statistic of § 6.2 now measures it post hoc.

Threshold.. The saturation threshold itself ($N_{50} \approx 51$, § 6.2) is estimated within this synthetic, static scenario family and should be read as family-specific rather than platform-independent: real request streams are heavy-tailed and arrive over time, and their effect on the threshold is untested (the priority weights are drawn uniformly on $\{1, \ldots, 10\}$, §5.4). The threshold is likewise unchanged under the bound/unbound split of § 6.2 (cap-free subset refit: midpoint identical, $n = 253$; the 47 ceiling-bound scenarios show no deficit).

Controls.. The hot-start control experiment (§ 6.6) indicates that G-BL initialization materially improves MOEA convergence; the coverage convergence reported here is conditional on this initialization, and full-scale random-init experiments are needed to establish the unconditional behavior. Similarly, the S1 coverage reversal under variant D (§ 6.5) now has a partial mechanistic account: the diagnostic arm of § 6.5 localizes the deficit to the quality content of f_2 (three objectives without it reach full coverage), leaving the excess-above-anchor part of the reversal to the crowding-distance account, which awaits verification with alternative crowding operators. The initialization and budget controls bound the findings above without fixing population values: their samples are small (10/4/5/5 scenarios with three seeds for the hot-start control; 15 runs per class for the random-init variant-D control), the scenarios are the first seeds of each density class in manifest order, and hot and random runs share scenario seeds, so every deficit is a within-seed paired difference expressed on the G-BL-normalized f_1^* scale, on which the greedy anchor reads exactly 1.0 by construction. The reference-point counts of MOEA-2 ($H = 13$) and MOEA-3 ($H = 91$) differ by construction, but a matched control at $p = 4$ ($H = 15$, §6.4) confirms the sparse-regime f_3 gain is not driven by reference-direction density.

Baselines.. Finally, we do not benchmark against learning-based schedulers (e.g., Chen et al. (2025)), which represent an alternative methodological family; a systematic comparison of physics-aware heuristic methods against data-driven approaches is left to future work.

8. Conclusion

We formulated single-satellite agile SAR scheduling as a three-objective problem in which coverage profit, geometric resolution, and NESZ radiometric quality are optimised together with NSGA-III, and we measured where the formulation pays for itself. Five configurations were compared on 200 scenarios across four density classes. In sparse request sets, re-optimising observation times buys large quality gains over the greedy coverage baseline at a stated coverage price: +40% in radiometric quality at a 15% coverage cost, +33.7% in geometric quality at 14%, and a controlled +4.9% third-objective margin under a matched radiometric-aware rule. Two measured boundaries delimit the gain. It contracts monotonically as the platform's squint envelope narrows, from +28% to +2.3% in geometric quality as the cap tightens from 45° to 15°. It vanishes at operational densities: beyond $N \approx 150$ the fronts we find converge to the greedy coverage anchor with quality margins below 2%. The vanished quality margins themselves are an effect of coverage pressure, which removes the freedom to move acquisitions, while solver convergence to lower-squint geometry aligns f_2 and f_3 through the shared $\cos\psi_{\mathrm{sq}}$ factor. The within-schedule trade-off persists at every density (negative correlation, §6.4), so the dense-regime convergence is an empirical effect of the schedule geometry, not a redundancy between objectives. Coverage convergence toward the greedy anchor is itself conditional on the hot start (§6.6), and GA-P-BL coincides with the greedy reference in all but one scenario, which delimits the benefit of adding physical objectives to profit-only search. For mission planning the rule is direct: quality-aware multi-objective scheduling pays on wide-envelope platforms with sparse request sets, and greedy methods attain the tested coverage reference at dense cadences. The squint-minimized heuristic matches or exceeds MOEA-3's radiometric quality at every density, and a knee-point schedule dominates the best placement-only mixture in 92.5% of scenarios for f_2, so the mixed family's advantage is an exchange-rate rather than a solver-ranking effect. The boundary estimates and the coupling decomposition give a baseline against which multi-satellite, multi-pass, and multi-mode extensions can be measured.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used AI-assisted tools (Claude, Codex) for manuscript preparation and revision assistance only. The author reviewed and edited all AI-assisted output and takes full responsibility for the content of this article.

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CRediT Authorship Contribution Statement

Liu Shuhao: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Resources, Data Curation, Writing – Original Draft, Writing – Review & Editing, Visualization.

Data Availability Statement

Solver code (NSGA-III via `pymoo` 0.6.1.6), scenario generators, fixed scenario seeds and configurations, processed JSON/CSV results (which pin the exact front realizations analyzed; within-instance solver process draws are not pinned), analysis scripts, and figures are available from the corresponding author. The companion repository is publicly available at `https://github.com/liushuhao/sar-quality-aware-scheduling`. Generated binary scenario files and per-run pickle objects are not tracked; they can be regenerated using the supplied scripts.

Declaration of Competing Interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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